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Review

Polymer microspheres in endovascular therapy: From fundamental research to clinical application



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Outline

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Highlights

- Comprehensive cross-disciplinary review of polymer microsphere embolization across five clinical specialties.
- Analysis links material properties of microspheres directly to clinical efficacy and safety outcomes.
- Biodegradable polymers like gelatin and chitosan show high promise for next-generation TACE and TARE.
- Microsphere embolization is effective for benign conditions: fibroids, BPH, hemorrhoids, and paragangliomas.

- Future directions include smart, imageable, and immunomodulatory embolic microsphere designs.

Abstract

This review highlights the current state of research in the field of the use of microspheres based on natural and synthetic polymers for embolization of hepatocellular carcinoma vessels, embolization of arteries in patients with paragangliomas, uterine myoma, prostatic hyperplasia and hemorrhoids. Particular attention in the review is paid to clinical studies of various sizes of microspheres to determine the effectiveness and safety of embolization processes. Based on the analysis of works over the past five years in the field of chemoembolization and radioembolization, it can be concluded that the most promising polymer matrices for the development of microspheres are cross-linked natural polymers, such as gelatin and chitosan. It has been established that polyvinyl alcohol microspheres and liquid embolizing agents based on ethylene-vinyl alcohol copolymer are effective and safe for preoperative embolization of patients with paragangliomas of various localizations. It is shown that the most promising embolizing agents for the treatment of uterine myoma, prostate hyperplasia and internal hemorrhoids are microspheres of cross-linked trisacryl coated with gelatin and microspheres based on polymethyl methacrylate coated with polyphosphazene. Unlike previous reviews focusing on single applications, this work provides a unique cross-disciplinary analysis of polymer microsphere embolization across multiple clinical specialties, highlighting material-design-clinical outcome relationships. The review may be of interest to bioengineers, materials scientists, chemists, and physicians working in the field of biopolymer materials and interventional therapies.

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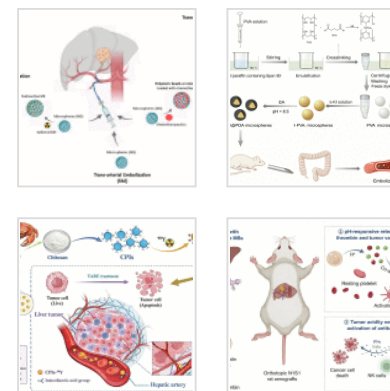
Polymer; Microspheres; Cancer; Artery; Chemoembolization; Radioembolization; Paraganglioma; Uterine fibroids; Prostate gland; Hemorrhoids

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1. Introduction

In recent years, new dosage forms providing prolonged and controlled release of drugs have been introduced into oncological practice. The necessity of application of such forms of delivery is connected to the fact that traditional dosage forms (tablets) provide a single, short-term and non-selective release of the drug. In turn, new drug delivery systems and dosage forms allow selective long-term release with a potentially lower level of side toxicity [1]. New dosage forms include primarily depot drug forms - modified forms for injection administration introducing therapeutic agent into the bloodstream at long intervals and accumulating in a certain diseased tissue environment [2].



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For the therapy of inoperable solid tumors (in particular, hepatocarcinoma), depot dosage forms are used, which are predominantly spherical in shape and have a diameter of 40 to 900 μm , depending on the diameter of the tumor microvascular bed. The use of microsized rather than nanosized forms is associated with the need for complete blocking (occlusion) of the tumor microvascular bed with subsequent ischemia and death of tumor tissue [3]. Microspheres, unlike microparticles of other shapes, can be made uniform in size, which facilitates their delivery to the tumor, and in some cases can be designed as porous or hollow, which allows for easy encapsulation of the chemotherapeutic [4].

Microspheres loaded with chemotherapy drugs by ion exchange or absorption improve the response rate of the loaded molecule [5], provide a slow controlled release, reducing the amount of drug freely moving to unwanted areas. Thus, it is possible to minimize the side effects of the cancer therapy [6]. In addition, the advantage of microspheres compared with the traditional chemotherapy method is that multiple doses of therapeutic can be administered with a single injection [7].

However, a problem with the delivery of microspheres to the tumor is the high density of some microspheres (in particular, the density of glass microspheres is 3.6 g/mL, which is approximately three times higher than the density of blood plasma) [8]. This can lead to rapid sedimentation and unintended embolization due to the difficulty of controlling the distribution of the microspheres in the tumor vessels. Other problems include: 1) non-biodegradability of some microspheres, such as the yttrium-90 radiolabeled TheraSphere glass microspheres (MDS Nordion, Ottawa, Canada) based on alumina and silica, which leads to their accumulation in the liver and complicates repeat embolization with repeated administration; 2) the inability to visualize the preparation in vivo, hindering accurate therapeutic assessment.

Currently, there is interest in radionuclide-labeled or chemotherapeutically immobilized microspheres based on cross-linked biocompatible and biodegradable polymers with a density comparable to blood, uniform particle size distribution, and sufficient elasticity

for effective delivery into arteries through a catheter and prevention of catheter occlusion [9].

In addition to the treatment of inoperable cancer, embolizing microspheres can be used to treat benign tumors and internal hemorrhoids. The advantage of embolization over surgery for these pathologies is a lower risk of damage to a non-target vessel due to precise occlusion of the target blood vessel, and a reduction in the postoperative risks of trauma, pain, and bleeding [10]. In patients with hemorrhoids, microspheres have fewer complications than liquid embolizers based on gel-forming polymers and are able to occlude further than microspirals [11].

To date, review articles have been published devoted to the use of microspheres for embolization in malignant liver tumors [[12], [13], [14], [15]]. Much attention in these works is paid to the side effects of therapy and the polymers used to manufacture the microspheres. For example, a review article published in 2022 [16] examined embolic agents based on polyvinyl alcohol (PVA), gelatin, polyacrylates, liquid embolics with controlled occlusive properties, presented the advantages and disadvantages of embolics, limitations and problems of developing embolic agents.

While previous reviews have predominantly focused on single applications or material classes, this review aims to fill this gap by providing a unique, integrated analysis of polymer microsphere applications spanning five distinct clinical specialties: oncology (hepatocellular carcinoma), vascular surgery (paragangliomas), gynecology (uterine fibroids), urology (benign prostatic hyperplasia), and proctology (hemorrhoids) (Fig. 1). We integrate findings from the past years to draw parallels and contrasts between these applications, evaluate how material properties and microsphere design dictate clinical utility in different vascular environments, and identify unifying principles for the rational design of next-generation embolic agents. By bridging materials science with diverse clinical practices, this work offers a unique, holistic perspective currently lacking in the literature, highlighting both the versatility and the specific requirements of microsphere-based embolization across the human vasculature.



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Fig. 1. Application of microspheres in various fields of medicine.

2. Microspheres for the treatment of hepatocellular carcinoma

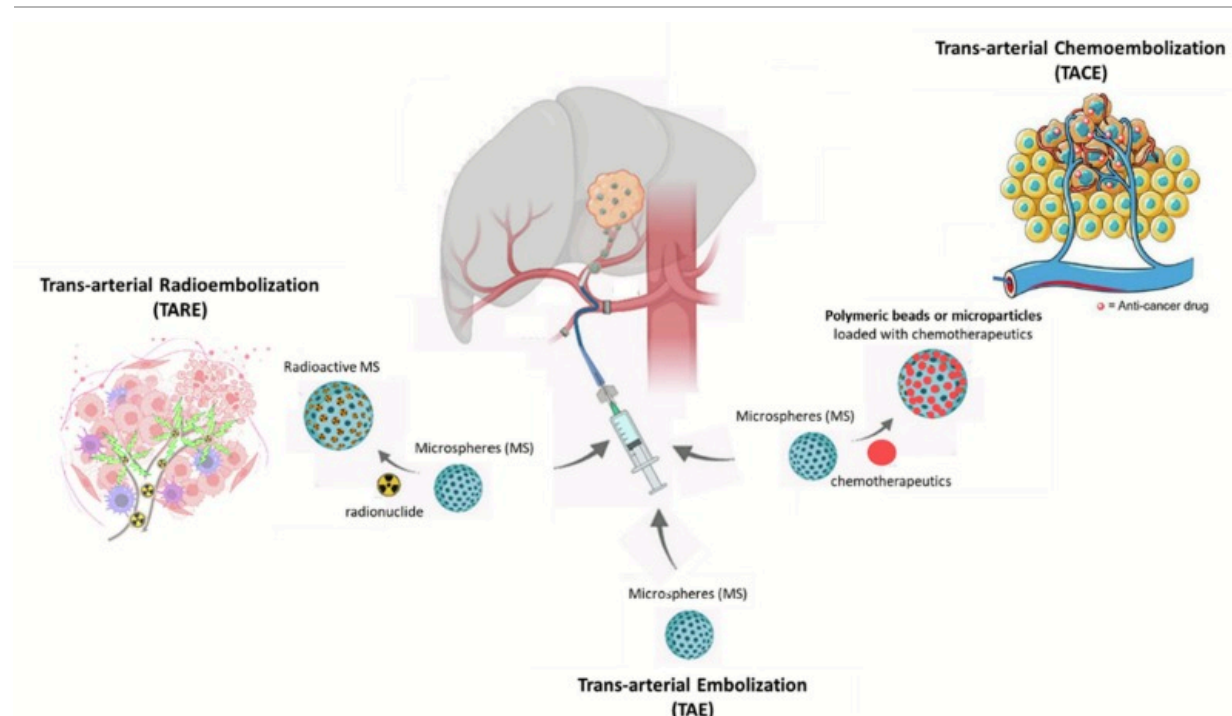
The liver is the parenchymatous organ most frequently affected by malignant neoplasms. Among all primary malignant neoplasms of the liver, the incidence of hepatocellular carcinoma (HCC) reaches 90% [17], while cholangiocarcinoma is only 10%. HCC is a significant global health problem, ranking third in mortality among oncological diseases [18].

The gold standard for treating patients with HCC is liver resection and, in rarer cases, liver transplantation. However, 50–70% of patients experience relapses within 5 years after radical resection [19]. In turn, in patients with HCC diagnosed at a late stage, with a heavy tumor load, surgical intervention is impossible.

Non-surgical treatments for HCC, such as chemotherapy with doxorubicin, oxaliplatin and tyrosine kinase inhibitors (sorafenib, lenvatinib), immunotherapy with immune

checkpoint inhibitors (atezolizumab in combination with bevacizumab) can only prolong the survival of cancer patients with subsequent development of resistance in 90% of cases [20]. Mutation of the TP53 tumor suppressor gene in patients with HCC also prevents improvement in patient survival with chemotherapy [21]. In addition, chemotherapy leads to side effects (hypertension, diarrhea, decreased appetite, weight loss) and serious adverse events (proteinuria) due to drug toxicity, which worsens the quality of life of cancer patients [22].

A promising replacement for traditional chemotherapy for the treatment of HCC is therapy using endovascular methods such as transarterial chemoembolization (TACE) and transarterial radioembolization (TARE) (Fig. 2).



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Fig. 2. Embolic microspheres for trans-arterial chemoembolization (TACE) and trans-arterial radio-

embolization (TARE). (Reproduced with permission from ref. [14]. Copyright 2024, American Chemical Society).

2.1. Transarterial chemoembolization (TACE)

The main types of TACE include oil chemoembolization and microsphere chemoembolization. Oil chemoembolization involves the delivery of an emulsion of chemotherapy drugs in an iodinated oily liquid based on poppy seed fatty acid esters (lipiodol) through a catheter into the arteries supplying the tumor area with subsequent tumor necrosis.

Oil embolization agents such as lipiodol do not cause irreversible or prolonged vascular occlusion but allow maintaining high concentrations of the chemotherapeutic agent in the tumor for several hours. However, due to the comparability of the chemotherapeutic agent concentration in the systemic circulation with intra-arterial administration, side effects occur, including embolization of normal liver tissue, penetration of the embolization agent into non-target organs [23]. Severe complications after oil chemoembolization are also possible, namely: liver abscesses, liver failure, and pseudoaneurysm formation [24]. Also, serious disadvantages of oil chemoembolization include the short-term (4–6h) stability of the chemotherapeutic agent emulsion in the oil embolization preparation.

An attractive safe alternative to oil chemoembolization, which allows increasing the interaction time of the chemotherapy drug with tumor cells (from 1 week to 2 months depending on the size) and avoiding the above-mentioned side effects, is the TACE method using loaded embolizing microspheres instead of a drug emulsion (DEB-TACE method) [[25], [26], [27], [28]]. The essence of this method consists of sorption of the chemotherapy drug by microspheres (in the volume or on the surface of the microspheres) with subsequent local introduction of microspheres saturated with the

chemotherapy drug using a catheter inserted directly into the hepatic artery feeding the tumor parenchyma.

Microspheres are mainly intended for the delivery of alkylating cytotoxic drugs (cisplatin, oxaliplatin) [29], which disrupt the integrity of DNA and block the processes of transcription and mitosis, and antitumor antibiotics or anthracyclines (doxorubicin [30], epirubicin [31], pirarubicin [32], idarubicin [33]), which promote partial cleavage of the DNA double helix and initiate the formation of single- and double-strand breaks.

To date, four types of cytostatic-loadable microspheres have been approved by the US Food and Drug Administration (FDA) for use in the DEB-TACE process (Table 1). These include the trisacryl-gelatin HepaSphere microspheres from Biosphere Medical, the Embozene microspheres based on polymethyl methacrylate coated with poly[bis(trifluoroethoxy)]phosphazene produced by CeloNova Biosciences, and the BC Bead and Callisphere microspheres based on sulfonated polyvinyl alcohol. Microspheres currently undergoing clinical trials include the LifePearl microspheres (MicroVention Europe, France) based on polyethylene glycol and the DEBDOX microspheres based on polyvinyl alcohol (Biocompatibles Ltd., UK). Embocept S starch-based microspheres are included in the European CIRSE therapeutic use guideline. In turn, MegaSphere polyvinyl alcohol-based microspheres are undergoing preclinical trials in China.

Table 1. Microspheres for DEB-TACE.

Product	Composition, size	Chemotherapy drug	Approval
HepaSphere (Biosphere Medical, Rockland, MA, USA)	Trisacryl-gelatin microspheres Size: 50–100, 100–150, 150–200µm	Epirubicin [31], doxorubicin [34]	FDA (USA) approved
Embozene Color-Advanced	Polymethyl methacrylate microspheres coated with	Doxorubicin [35]	FDA (USA) approved

Product	Composition, size	Chemotherapy drug	Approval
Microspheres (CeloNova BioSciences, Newnan, Georgia, USA)	polyphosphazene (Polyzene®-F, poly[bis(trifluoroethoxy) phosphazene]) Size: 40, 100, 250, 400, 500, 700, 900µm		
DC Bead (Biocompatibles Ltd., UK)	Microspheres based on sulfonated polyvinyl alcohol Size: 70–150, 100–300, 300–500, 500–700µm	Doxorubicin [36]	FDA (USA) approved
Callisphere (Jiangsu Hengrui Medicine Co. Ltd., Jiangsu, China)	Microspheres based on sulfonated PVA crosslinked with glutaraldehyde Size: 100–300, 300–500µm	Doxorubicin [30], epirubicin [37], pirarubicin [38], oxaliplatin [29]	Institutional Review Board of Medical Centers of China and FDA (USA) approved
DEBDOX (Biocompatibles Ltd., UK)	Hydrogel microspheres based on polyvinyl alcohol Size: 70–150, 100–300, 300–500, 500–700µm	Doxorubicin [39]	Undergoing clinical trials
LifePearl (MicroVention Europe, Saint-Germain-en-Laye, France)	Polyethyleneglycol based microspheres Size: 100±25, 200±50, 400±50µm	Doxorubicin [40], epirubicin [41], idarubicin [33]	Undergoing clinical trials

Product	Composition, size	Chemotherapy drug	Approval
Embocept S (PharmaCept GmbH, Berlin, Germany)	Starch microspheres from partly hydrolyzed starch, cross-linked and substituted with glycerol ether groups, Size: 50µm	Doxorubicin, oxaliplatin [42]	Included in the European therapeutic use guidelines CIRSE
MegaSphere (Suzhou Hengrui Garishen Biomedical Technology Co., Ltd., Jiangsu, China)	Polyvinyl alcohol based microspheres Size: 100–300µm	Doxorubicin	Undergoing preclinical trials

As can be seen from the data in [Table 1](#), doxorubicin is most often included in the operational documentation for the use of cytostatically loaded microspheres. This drug is well compatible with polymeric materials, is effectively immobilized by negatively charged polymer emboli and is released in a modified manner at the site of occlusion.

To obtain microspheres loaded with cytostatics, the method of solvent evaporation from the emulsion and the method of spray drying are mainly used [43], as well as microfluidics technology. The emulsion method of microsphere synthesis consists of preparing aqueous and organic phases containing polymer and drug with their subsequent emulsification during mixing to form a “water-in-oil” emulsion. The volatile solvent is evaporated either at room temperature or under reduced pressure. After evaporation of the solvent, the emulsion microdroplets harden to form solid microspheres, which are filtered and dried. Ultrasonic treatment is also used to accelerate the process of solvent evaporation during the synthesis of microspheres with a small diameter of emulsion droplets.

The essence of the spray drying method consists of dispersing a solid cytostatic in a polymer solution with subsequent feeding of the mixture through a sprayer into a drying chamber [44]. In the chamber, heated air dries the particles and transports them to a jet separator for collection. This method allows instantaneous evaporation of the solvent and encapsulation of both water-soluble and hydrophobic cytostatics.

Microfluidic technology involves the use of a microfluidic droplet generator with a T-junction, consisting of two gas-tight syringes, and two programmable syringe micropumps. To prevent droplet merging, a DC motor is used, attached to an inverted funnel and stirring the collector bath. Microspheres are formed from microdroplets either in a collector bath [45] or when moving in a serpentine channel after the T-junction [46]. Microfluidic technology allows for high-precision control of liquid flow and the production of uniform-sized and smooth microspheres [46].

To achieve effective occlusion of peripheral tumor arteries, which large-diameter microspheres (more than 180–200 μm) cannot penetrate, liquid embolic agents are often used that are capable of rapidly diffusing into peripheral tumor arteries due to their fluidity and changing from a liquid to a gel state at body temperature in situ. This allows for effective occlusion of multi-branching tumor blood vessels. Such liquid embolic agents are made on the basis of poly(N-isopropylacrylamide), poly(N-isopropylacrylamide-co-butyl methacrylate) and poly(N-isopropylacrylamide-b-2-(diethylamino)ethyl methacrylate) block polymer, poloxamer 407 [[47], [48], [49]]. For example, the authors of [49] showed that a thermosensitive nanogel based on poloxamer 407, which is a triblock copolymer with a central hydrophobic chain of polyoxypropylene and two identical side hydrophilic chains of polyoxyethylene and loaded with doxorubicin and selenium nanoparticles, has a high efficiency of occlusion of both main and peripheral vessels of HCC. Also, the liquid embolic agent reduced the tumor volume in a HepG2 mouse xenograft model in vivo.

The clinical efficacy of DEB-TACE is linked to the material properties of the microspheres. For example, the ionic binding capacity of sulfonated PVA microspheres (DC Bead,

CalliSpheres) allows controlled and sustained release of positively charged chemotherapeutic agents like doxorubicin, minimizing systemic exposure and toxicity. Vice versa, the porous structure of trisacryl-gelatin microspheres (HepaSphere) enables high drug loading and a slower release profile, which may prolong the local therapeutic effect. The size and elasticity of microspheres are critical for distal penetration into the tumor vasculature. Smaller (70–150 μ m) and more deformable microspheres can occlude deeper and narrower tumor vessels, leading to more extensive ischemia and higher tumor response rates, as evidenced by clinical studies [50,51]. Moreover, the non-biodegradable nature of PVA and polymethyl methacrylate microspheres provides permanent occlusion, which is desirable for long-term vessel blockade, but may complicate retreatment due to persistent accumulation in the liver.

Another promising minimally invasive treatment for HCC, alternative to TACE is transarterial radioembolization (TARE) which will be considered further.

2.2. Transarterial radioembolization (TARE)

Transarterial radioembolization (TARE) is a relatively new method of intravascular interventions that combines vessel embolization and tumor irradiation with radioactive isotopes included in glass or resin spherical microparticles. Unlike external or distance radiotherapy, in which radiation is delivered to the tumor from the outside, transarterial radioembolization provides selective delivery of particles containing radioactive sources (yttrium-90, holmium-166) into the vessels supplying the tumor and its metastases. Microspheres effectively penetrate deep into the tumor, delivering therapeutic radiation and minimizing harm to surrounding healthy tissues [52].

TheraSphere glass microspheres and resin microspheres SIR-Spheres, both containing yttrium-90 as a radionuclide, have been FDA approved for use in TARE treatment in patients with HCC and liver metastases [53,54]. In April 2015, the European Medicines Agency (EMA) approved poly-L-lactide microspheres QuiremSpheres containing holmium-166 as a radionuclide (Table 2) [55].

Table 2. Microspheres for TARE in patients with HCC and liver metastases.

Product	Composition, size and activity	Radioactive isotope	Approval
TheraSphere (MDS Nordion, Ottawa, Canada)	Glass microspheres, diameter of 20–30 μ m and an activity of 2500Bq per sphere at calibration	Yttrium-90	FDA (USA) approved
SIR-Spheres (Sirtex Medical Ltd., North Sydney, Australia)	Microsphere based on cation exchange polystyrene resin (Bio-Rad, Hercules, CA) with Y90 impregnated on the surface of the sphere, a sphere diameter of 20–60 μ m, and an activity of 50Bq per sphere at calibration	Yttrium-90	FDA (USA) approved
QuiremSpheres (Quirem Medical BV, Deventer, the Netherlands)	Microspheres made from poly-l-lactide, average diameter 30 μ m (range of 15–60 μ m) and activity of 80Bq per sphere at calibration	Holmium-166	EMA (EU) approved

Microspheres for TARE are synthesized by solvent evaporation method. In particular, to obtain QuiremSpheres microspheres, a solution of holmium acetyl acetonate and poly-l-lactide in chloroform and a solution of PVA in water are prepared. The organic solution is then added to the aqueous solution with stirring and the chloroform is evaporated. After washing the microspheres with 0.1 N hydrochloric acid and water, they are sieved and dried in a nitrogen atmosphere [56].

The advantages of QuiremSpheres microspheres are: 1) a shorter half-life of holmium-166 (26.8h) compared to yttrium-90 (64.1 h), which should lead to an increase in the radiobiological effect due to the high initial dose rate [57]; 2) the ease of nuclear imaging of QuiremSpheres to determine their distribution in the liver, surrounding tissues and

lungs due to the strong paramagnetism of holmium and the emission of low-energy (81 keV) gamma photons [58].

DEB-TACE has advantages over TARE in terms of survival and side effects. A 2025 clinical trial involving 279 patients in Saudi Arabia showed that patients who underwent DEB-TACE had longer survival compared to TARE. It was also found that patients with TARE were more likely to experience liver complications such as liver decompensation. However, this study was performed in patients with early to intermediate stage HCC. Further randomized clinical trials are needed in patients with higher tumor burden and patients who are not suitable for DEB-TACE [59].

In addition, the multifaceted nature of the therapy and the need for participation of physicians of various specialties in the treatment of patients are the main obstacles (along with the high cost of TARE) to the widespread implementation of this method. The TARE method requires multidisciplinary cooperation of specialists from various fields - interventional radiology, radiation physics, medical and surgical oncology, hepatology and nuclear medicine - to assess the condition of patients and select the optimal treatment method. Such a decision is made only after a joint assessment of the physicians examining the tumors, including various functional areas. TARE is performed by a specially trained interventional radiologist specializing in this field. It is also necessary to ensure control over the technique of performing the procedures, the work of the entire surgical team, including the interventional radiologist, anesthesiologist, radiologist, radiation physics specialist, interventional technologist, nursing staff and radioisotope technologists.

The choice of matrix material for TARE microspheres determines not only the delivery and retention of radionuclides but also the post-embolization inflammatory response and long-term safety. Glass microspheres (TheraSphere) are non-biodegradable and have a higher density, which may lead to faster sedimentation and heterogeneous distribution in the tumor vasculature. In contrast, resin-based microspheres (SIR-Spheres) are lighter and biodegradable, allowing more uniform distribution and eventual clearance, although

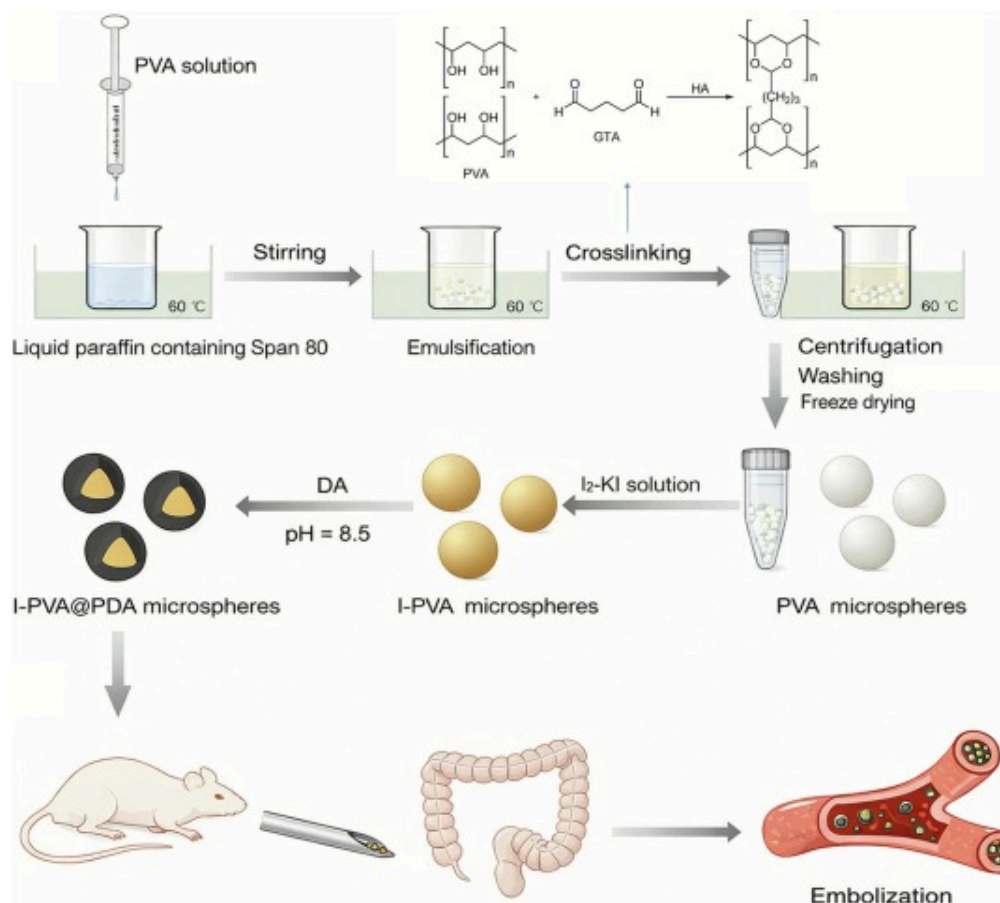
with a potential for late recanalization. The recent development of biodegradable polymer microspheres (poly-l-lactide QuiremSpheres) combines the advantages of controlled degradation, which avoids permanent foreign body retention, with the ability to carry radionuclides like holmium-166 that offer easier imaging and a shorter half-life [60]. The material properties thus directly impact the radiation dose distribution, the risk of non-target irradiation, and the feasibility of repeat treatments.

As we can see, the properties of polymer matrices are of great importance for studying the shortcomings of approved microspheres and creating new products, studying the effectiveness of treatment, patient survival, and side effects. Therefore, we further considered some polymer matrices used to manufacture new microspheres.

3. Polymer matrices for the production of microspheres

3.1. Polyvinyl alcohol

Polyvinyl alcohol (PVA) is a biocompatible polymer that has been used in biomedicine since the middle of the last century. PVA-based microspheres are capable of elastic deformation and are highly elastic, which will allow the lumen of the vessel to be completely filled at different levels during the delivery of microspheres through an arterial catheter and completely block its blood supply (Fig. 3) [61,62].



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Fig. 3. Polyvinyl alcohol (PVA) microspheres with Iodine (I) and polydopamine (PDA) for the enhanced embolization of liver portal vein. (Modified picture. Reproduced with permission from ref. [61]. Copyright 2022, Elsevier).

PVA-based microspheres are effective and safe for DEB-TACE in patients with HCC. PVA-based microspheres are negatively charged and bind positively charged chemotherapeutic agents through ionic interactions. In [63], glutaraldehyde-crosslinked sulfonated PVA Callisphere microspheres (size 100–300μm, Suzhou Hengrui Jialisheng Biomedical

Technology, China) loaded with cytostatic epirubicin were used for tumor vessel embolization in 125 patients with HCC. The microspheres provided permanent blockage of tumor vessels and occlusion of tumor blood supply. Callisphere microsphere embolization showed high efficiency with no possible deterioration of liver function and high patient survival.

In the work [64] it was determined that MegaSphere polyvinyl alcohol-based microspheres (100–300 μ m, Suzhou Hengrui Garishen Biomedical Technology Co., Ltd., Jiangsu, China) loaded with doxorubicin are effective in arterial embolization therapy of HCC due to their ability to elastically deform from spherical to ellipsoid or flattened structures and reach the lumens of distal branching vessels of the renal artery. A study of the release of doxorubicin from microspheres in 5% saline (pH7.4) showed that the concentration of doxorubicin was 100mg/g within 30min, which was 20% higher than that of CalliSphere microspheres. The prolonged release of the chemotherapeutic agent indicates a stable antitumor effect of this delivery system. Also in the work [64] an in vivo study was conducted on a VX2 rabbit liver tumor transplant model and an increased frequency of apoptosis was established after treatment with the DEB-TACE method using MegaSphere microspheres. It should be noted that tumor response rates are higher in HCC patients who undergo embolization with smaller microspheres (70–150 μ m in size) compared to larger ones (100–300 μ m).

In a retrospective study performed by Lee, B. C. et al. [65] involving 212 patients with tumors smaller than 3cm, it was found that with embolization with doxorubicin microspheres measuring 70–150 μ m, the objective tumor response rate to therapy was 91.7%, and with embolization with microspheres measuring 100–300 μ m – 84.5%. This is probably due to the fact that microspheres of smaller diameter penetrate deeper into tumor tissues compared to larger microspheres.

PVA-based microspheres of 70–150 μ m in size are not only more effective but also safer. In particular, DEBDOX microspheres loaded with doxorubicin (Biocompatibles Ltd., UK)

showed no hepatobiliary adverse events such as bile duct injury, biloma, liver abscess, and liver infarction [66].

In addition to chemotherapy drugs delivery, PVA-based microspheres can be used to deliver radionuclides. In [67], SIR-Spheres microspheres (Sirtex Medical Limited, Sydney, Australia) with a diameter of 20–60 μm were used to deliver yttrium-90 to tumors in patients with HCC with an average age of 70 years. The authors found a radiological response to treatment with SIR-Spheres microparticles in 27 patients. In another study [68], involving a larger number of patients ($n=36$), a 5-year survival rate of 89% was shown.

Considerable attention is also paid to porous PVA microspheres for controlled delivery of cytostatics. Porous PVA-based microspheres are obtained by emulsification followed by cross-linking and final lyophilization [69]. First, an aqueous solution of PVA is added dropwise to an organic phase containing n-heptane or liquid paraffin and Span80 (emulsifier). Then the resulting water-in-oil emulsion is cooled with liquid nitrogen and the organic solvent is removed. For additional stabilization of the microspheres, they are cured by chemical cross-linking using glutaraldehyde as a cross-linking agent [70]. The three-dimensional hierarchical architecture of porous microspheres facilitates higher percentage of drug loading. Porous microspheres based on PVA cross-linked with glutaraldehyde and functionalized with phenyl sulfonate groups bind positively charged amino groups of doxorubicin via SO_3^- groups both on the surface and in the volume of microspheres [70]. Doxorubicin loading efficiency using these microspheres was 82.5% after one hour of immersion in a doxorubicin solution with a concentration of 2 mg/mL, while for non-porous particles it was 56.9%.

In our opinion, the disadvantages of PVA-based embolizing microspheres are the technological difficulties in obtaining particles of uniform size, the high coefficient of friction of microparticles and their tendency to aggregation [71]. The latter factor can lead to a change in the size of the microspheres, difficulty in their passage through the catheter and occlusion of large vessels. In addition, PVA-based microspheres are

considered as non-biodegradable, which can lead to thrombosis and non-target damage [72].

The clinical performance of PVA-based microspheres is a direct consequence of their tunable material properties. By altering the degree of sulfonation and cross-linking, the drug loading capacity and release kinetics can be modulated to match the specific pharmacokinetic requirements of different chemotherapeutic agents. The elasticity of PVA microspheres, which is higher than that of glass or resin particles, allows them to deform and navigate through tortuous vasculature, leading to more distal and complete occlusion. However, the non-biodegradability of PVA may lead to long-term foreign body reactions and permanent vessel occlusion, which, while beneficial for sustained embolization, can hinder future interventions and cause chronic inflammation in some patients.

3.2. Chitosan

Chitosan, which is a common natural polysaccharide biopolymer, attracts the attention of researchers in the field of biomedicine due to its non-toxicity, easy availability, biodegradability, low immunogenicity, and antimicrobial properties [[73], [74], [75]]. For the application of chitosan in TARE, it needs to be functionalized because chitosan has low radioisotope labeling efficiency and poor adsorption stability [76]. Poor adsorption stability can lead to significant amounts of radioisotope from the surface of the microspheres entering the bloodstream and causing radiation damage to liver tissue [77].

In [78], microspheres based on chitosan functionalized with polyglycidyl methacrylate and iminodiacetic acid (Fig. 3) were obtained in three steps. At the first stage, a water-in-oil emulsion was prepared by the membrane emulsification method, which consisted of passing the dispersed phase (acetic acid solution of chitosan) through homogeneous membrane pores into the continuous phase (petroleum ether and Span80 paraffin liquid) under the action of nitrogen. After emulsification, the functionalized chitosan was cross-linked with glutaraldehyde. At the third stage, chitosan was functionalized with glycidyl

methacrylate using free-radical polymerization in an aqueous medium (initiating system potassium persulfate and ferrous sulfate) and heated in an alkaline solution of iminodiacetic acid. The resulted microspheres demonstrated good adsorption capacity associated with stable interaction of yttrium (III) and sodium carboxylate (Fig. 3). The radioembolizing microspheres labeled with yttrium-90 had a spherical shape, an average diameter of 20.9 μm and a density of 1.47g/cm³, close to the density of blood plasma. Biodegradability testing demonstrated degradation of the microparticles in blood serum after 6 weeks. Theoretical calculation of the radioactivity of the microspheres showed higher activity compared to commercial TheraSphere glass microspheres, SIR-Spheres resin microspheres and silica core/yttria shell ceramic microspheres. However, a disadvantage of the synthesized microspheres is the sensitivity of yttrium adsorption to the pH of the medium. At pH values 1 and 2, electrostatic repulsion was observed between the microspheres and yttrium Y³⁺, at pH 5–7, a negative charge was observed on the surface of the microspheres due to deprotonation and electrostatic attraction to yttrium Y³⁺. The optimal pH was 6. At pH over 7, a precipitate of yttrium hydroxide was formed.

The disadvantages of chitosan for the synthesis of microspheres also include the difficulty of creating a stable emulsion of small droplets in an acetic acid medium and obtaining uniform-sized microspheres. In addition, the use of chitosan microspheres in TARE in patients with HCC requires preliminary in vivo studies in mice and visualization of the distribution of yttrium-90 in the tumor using magnetic resonance imaging.

The functionalization of chitosan microspheres demonstrates how material engineering can overcome the natural limitations of chitosan, such as poor radioisotope labeling efficiency and adsorption stability. By introducing carboxylate groups, the microspheres gain a high affinity for yttrium-90, ensuring stable binding and minimal leakage of the radionuclide into the circulation [78]. This material modification is crucial for reducing radiation-induced damage to non-target tissues and achieving effective tumor irradiation. Furthermore, the biodegradability of chitosan allows eventual clearance of the

microspheres, which is advantageous for preventing long-term complications and enabling repeat embolization if needed.

3.3. Gelatin

Microspheres based on gelatin, a natural and biodegradable biopolymer, along with chitosan, have been used as a matrix for the fabrication of microspheres. For example, in [79], gelatin microspheres loaded with doxorubicin and the adenosine receptor antagonist A2AR SCH-58261 (4-(Furan-2-yl)-10-(2-phenylethyl)-3,5,6,8,10,11-hexazatricyclo[7.3.0.0]dodeca-1(9),2,4,7,11-pentaen-7-amine) were developed for the embolization of tumor vessels in hepatocellular carcinoma. For subsequent loading into gelatin microspheres, the A2AR antagonist, which is poorly soluble in water, was first loaded into lipid nanoparticles. Gelatin microspheres were obtained using the emulsion cross-linking technique. To prepare the aqueous phase, doxorubicin and gelatin were dissolved in a pre-prepared nanosuspension of SCH-58261 in lipid nanoparticles. The organic phase was prepared by dissolving Span80 in liquid paraffin. After mixing the two phases, a water-in-oil emulsion was formed, which was then placed in a bath with ice water and glutaraldehyde was added to the solidified emulsion. Gelatin microspheres were obtained after centrifugation, washing with isopropanol from residual organic solvents and air drying. The obtained microspheres showed a sustained release of doxorubicin and SCH-58261 for 30 days. It was also found that the microspheres not only provided complete occlusion and enhanced apoptosis of tumor cells, but also maturation of dendritic cells and antigen presentation. This enhanced tumor-specific T-cell immunity.

This strategy demonstrates approaches shift from purely occlusive and cytotoxic embolization toward immunomodulatory embolization. By locally releasing an A2AR antagonist, these microspheres counteract the adenosine-mediated immunosuppression prevalent in the hypoxic tumor microenvironment post-embolization. This not only promotes direct tumor cell apoptosis but also reverses T-cell exhaustion and enhances antigen presentation, thereby stimulating a systemic anti-tumor immune response. This

approach synergizes with emerging immunotherapies and addresses a key limitation of conventional TACE – the worsening of the immunosuppressive state which can contribute to relapse and metastasis.

In addition to adenosine receptor antagonists, microspheres based on gelatin cross-linked with glutaraldehyde were loaded with doxorubicin and magnetite nanoparticles (Fe_3O_4 NPs) [80]. The diameter of the microspheres was 200–300 μm . The addition of magnetite nanoparticles to the microspheres contributed to an increase in the antitumor efficacy of doxorubicin when heated in a microwave oven due to the absorption of microwave radiation by the nanoparticles and its conversion into thermal energy, as well as the activation of such a type of controlled cell death as ferroptosis. Improvement of the antitumor effect was evidenced by a decrease in the ability of LM3 hepatoma cancer cells to proliferate. In turn, an in vivo study on the VX2 rabbit liver tumor model confirmed that microsphere embolization combined with microwave irradiation can provide an excellent therapeutic effect without serious side effects such as cardiomyopathy.

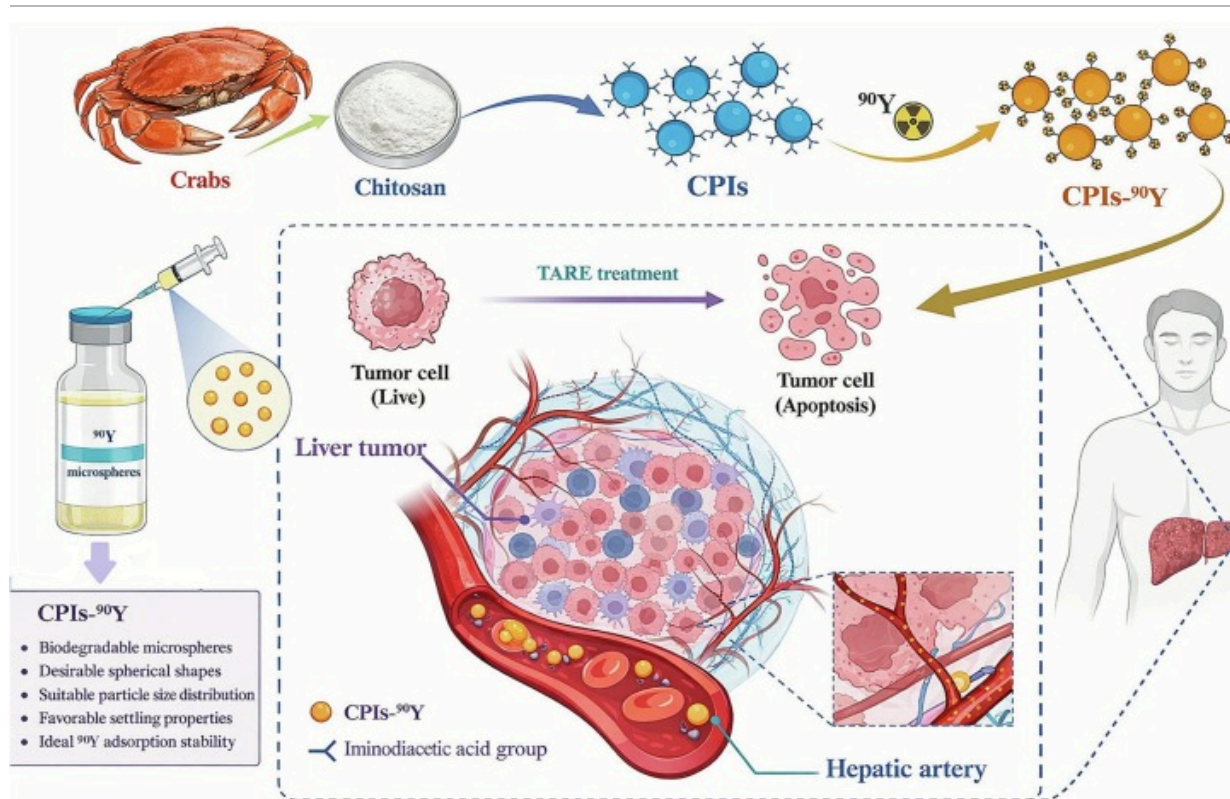
Thus, gelatin microspheres loaded with drugs specifically aimed at different targets allow DEB-TACE to be combined with immunotherapy and hyperthermia therapy. However, due to its unpredictable biodegradability, gelatin-based embolization may result in early restoration of blood flow. Thus, gelatin is not suitable for permanent occlusion. To ensure complete occlusion of tumor blood vessels, thrombin and calcium carbonate nanoparticles are loaded into gelatin microspheres, promoting tumor acidity-sensitive release and the formation of intravascular blood clots (Fig. 5) [81].

The versatility of gelatin as a matrix material allows incorporation of various therapeutic agents and even nanoparticles, enabling combination therapies that address multiple aspects of tumor biology. For example, the addition of magnetite nanoparticles (Fe_3O_4) to gelatin microspheres introduces the capacity for magnetic hyperthermia and ferroptosis induction, thereby enhancing the antitumor effect of the loaded chemotherapeutic [80]. The biodegradability of gelatin can be both an advantage and a disadvantage: while it allows eventual clearance of the embolic agent, it may also lead to premature

recanalization if the degradation rate is not matched with the time required for complete tumor necrosis. Therefore, controlling the cross-linking density of gelatin is essential to adjust the degradation profile to the clinical need.

3.4. Hyaluronic acid

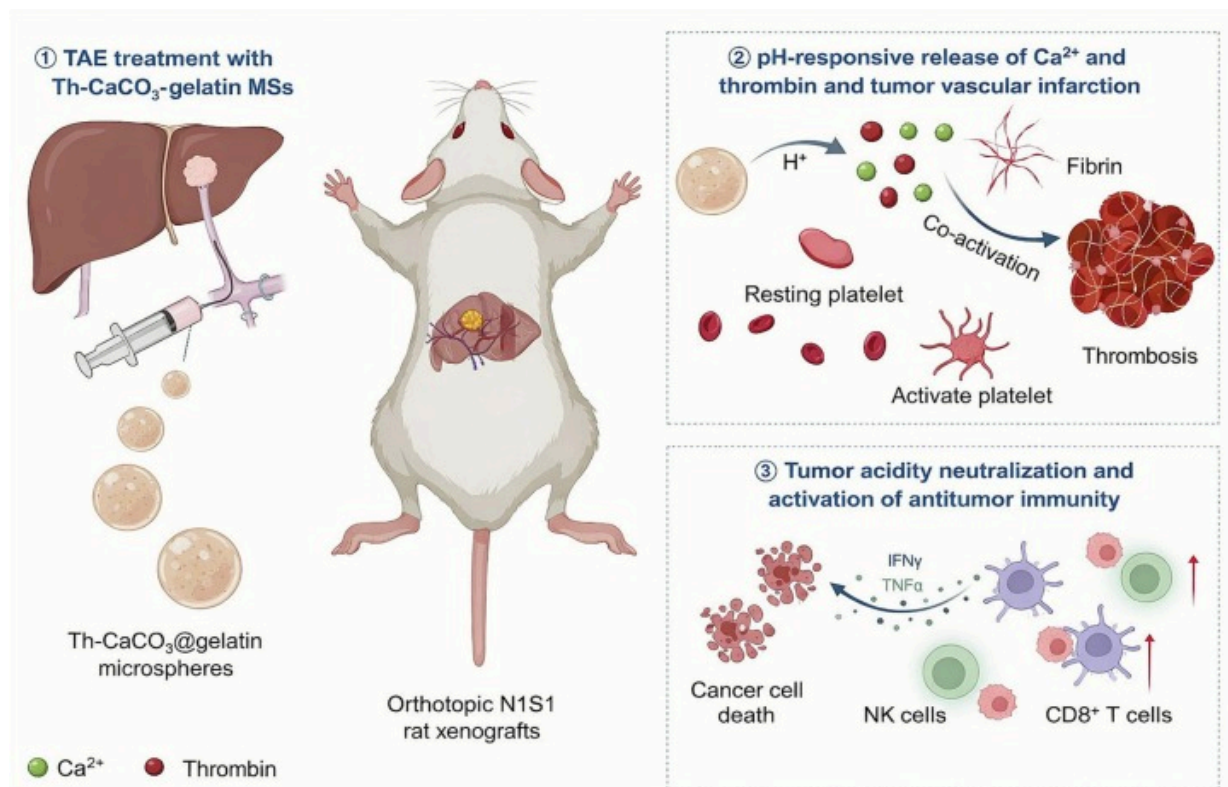
Hyaluronic acid (HA) is a natural hydrophilic polysaccharide contained in the extracellular matrix and possessing biocompatibility and biodegradability ([Fig. 6](#)) [82]. The biodegradability of HA is due to the presence of the enzyme hyaluronidase in it [83]. These properties of HA make its use promising for the production of embolic microspheres. In particular, in the work [84] using droplet microfluidic technology, monodisperse microspheres (average diameters of microspheres 154, 182, 209, 250, 290, 337 μm) based on HA and adipic acid dihydrazide as a cross-linking agent ([Fig. 4](#)) were obtained. An aqueous solution of HA and adipic dihydrazide was the dispersed phase, and the continuous phase was n-octanol containing 5% Span80 (surfactant).



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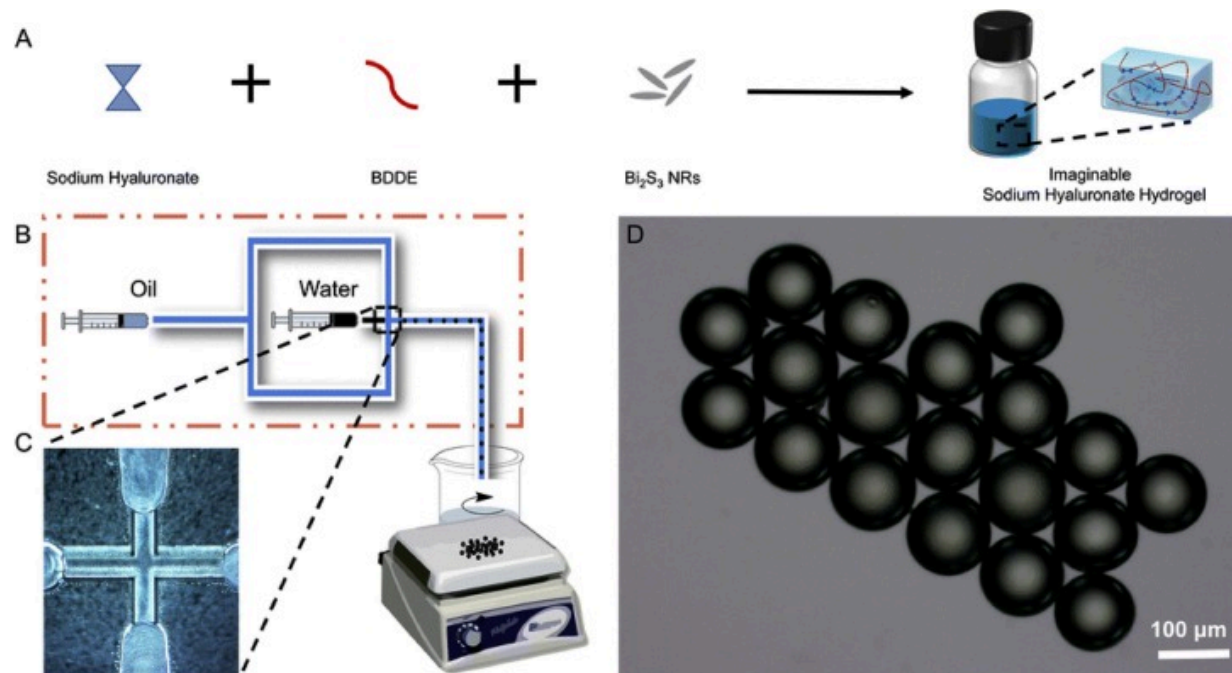
Fig. 4. Chitosan-based microspheres (CPI) as biodegradable yttrium-90 delivery system for radioembolization therapy. (Modified picture. Reproduced with permission from ref. [78]. Copyright 2025, Elsevier)..



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Fig. 5. Thrombin-loaded calcium carbonate (CaCO₃) nanoparticle-embedded gelatin procoagulant embolic microspheres (Th-CaCO₃@gelatin MSs) prepared via a microfluidic process. (Modified picture. Reproduced with permission from ref. [81]. Copyright 2024, Elsevier).



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Fig. 6. Microfluidic fabrication of X-ray-visible sodium hyaluronate microspheres for embolization. (A) Imaginable sodium hyaluronate (SH) gel preparation. (B) Scheme of microspheres fabrication. (C) Local magnification image of a flow-focused droplet microfluidic chip. (D) Optical micrograph of hydrated SH microspheres. (Reproduced with permission from ref. [82]. Copyright 2023, RSC published under a Creative Commons Attribution-NonCommercial 3.0 Unported License).

The obtained microspheres did not collapse after compression to 65% and were restored to their original morphology within 10s, which indicates their excellent elastic properties. The study of the biodegradability of the microspheres showed that at a mass ratio of adipic acid dihydrazide and HA of 0.05g/g, the microspheres dissolve in a phosphate buffer and completely disappear on the 5th day of observation. In addition, the survival of mouse fibroblasts L929 during their joint cultivation with a suspension of microspheres

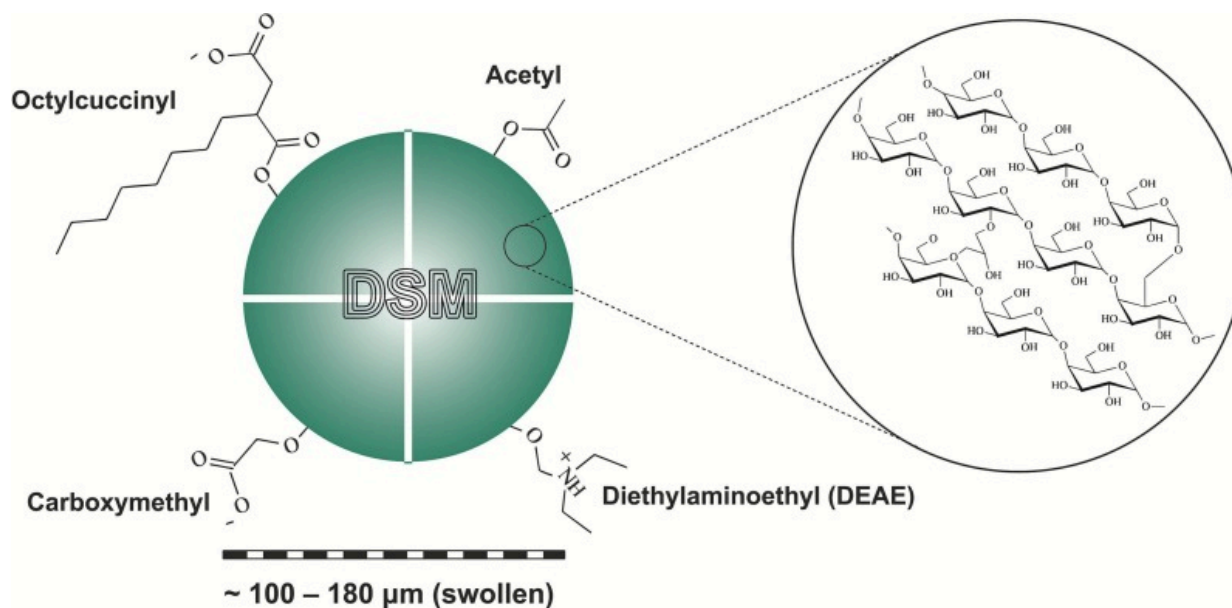
for 72h exceeded 88%, and the hemolysis rate of rabbit erythrocytes was less than 2%. The presented data corresponded to the biodegradability and biocompatibility of the microspheres and met the medical and biological requirements [84].

Furthermore, an in vivo study of HA-based microspheres functionalized with 2-aminoethyl methacrylate (AMA) demonstrated effective blockade of the blood supply to rabbit liver tumors for more than one month. HA-AMA microspheres (90 μ m) loaded with epirubicin contributed to significantly better suppression of HCC growth in rabbits and a lower number of metastatic nodules in the lungs of rabbits at day 14 compared to Callispheres microspheres (100–300 μ m) loaded with epirubicin. In turn, immunohistochemical analysis demonstrated a decrease in staining of the integral cellular glycoprotein CD44 on the tumor vasculature after six hours of microsphere embolization due to the binding of the microspheres to CD44-positive endothelial cells in the tumor vasculature [85].

The exceptional elasticity and rapid recovery of HA-based microspheres, derived from their hydrophilic and cross-linked network, enable them to withstand the hemodynamic forces in the vasculature and maintain occlusion over time. Moreover, the ability of HA to bind to CD44 receptors overexpressed on tumor endothelial cells provides an active targeting mechanism, enhancing the specificity and retention of the microspheres within the tumor vasculature. This targeting effect not only improves the efficiency of vessel occlusion but also increases the local concentration of any loaded drug, thereby potentiating the therapeutic outcome. The biodegradability of HA via hyaluronidase ensures that the microspheres are eventually cleared, reducing the risk of long-term complications. Thus, HA-based microspheres show promising therapeutic application in the embolization of HCC vessels. However, clinical studies of hyaluronic acid-based microspheres are currently lacking.

3.5. Starch

Degradable starch microspheres (DSM) are composed of starch chains cross-linked by glycerol ether groups and are readily degraded by blood α -amylase, with a half-life of approximately 35–50 min both in vitro and in vivo experiments [86]. Degradable starch microspheres (DSMs) are starch chains cross-linked with epichlorohydrin, forming glycerol–ether links. DSMs have been used for many years for temporary vascular occlusion and drug delivery in treatment of malignancies (Fig. 7) [87].



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Fig. 7. Schematic drawing of starch degradable microsphere (DSM) and their possible modifications.

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The use of DSM as embolic agents has only recently (2021) gained wider clinical application in Europe. Commercially produced starch-based microspheres EmboCept S (PharmaCept, Berlin, Germany) preserve liver function and are well tolerated by patients. EmboCept S microspheres are prepared in a water-in-oil emulsion in the presence of a

cross-linking agent, followed by washing with an organic solvent and air drying. The cross-linking reaction occurs at a high concentration of sodium hydroxide to deprotonate the starch hydroxyl groups [88].

DEB-TACE using doxorubicin-loaded DSM can reduce the systemic inflammatory response index in patients with HCC [89] and provide longer patient survival, good tumor response rate, and low incidence of treatment-related adverse events [90]. In the work [90], a clinical study is reported involving 79 patients with inoperable and recurrent HCC. EmboCept S microspheres loaded with doxorubicin served as the embolizing agent. As a result, the authors found that the only adverse effects were biological ones associated with an increase in bilirubin levels, which were eliminated by medication. Thus, DEB-TACE using DSM is a universal method with a favorable safety profile in terms of treatment-related adverse effects. In the future, it is recommended to conduct a combined study that includes immunotherapy in addition to chemotherapy.

The rapid biodegradability of starch microspheres by α -amylase results in a temporary occlusion, which is particularly useful in clinical scenarios where only short-term vessel blockade is desired, such as in the prevention of systemic toxicity during chemotherapy infusion [86]. However, for long-term tumor control, the quick degradation of DSM may be a drawback, as it can allow early revascularization. Therefore, DSM are often used in combination with other embolic agents or for drug delivery in a controlled, short-term manner. The clinical success of DSM in reducing systemic inflammatory response and improving patient survival highlights the importance of matching the degradation rate of the embolic material to the therapeutic goal. The development of DSM underscores the trend toward tunable, transient embolic agents designed for specific pharmacokinetic and hemodynamic windows, contrasting with permanent occlusive materials.

3.6. Zein

Zein, a major storage protein derived from corn, represents a promising but less conventional natural biopolymer for embolization applications. Its unique solubility

profile (soluble in aqueous ethanol but insoluble in pure water) allows it to be formulated as injectable liquid embolics that precipitate upon contact with blood, forming stable occlusions [91]. This property, combined with its biodegradability, biocompatibility, and ability to encapsulate drugs, gained interest in zein as a versatile platform for both solid and liquid embolic agents.

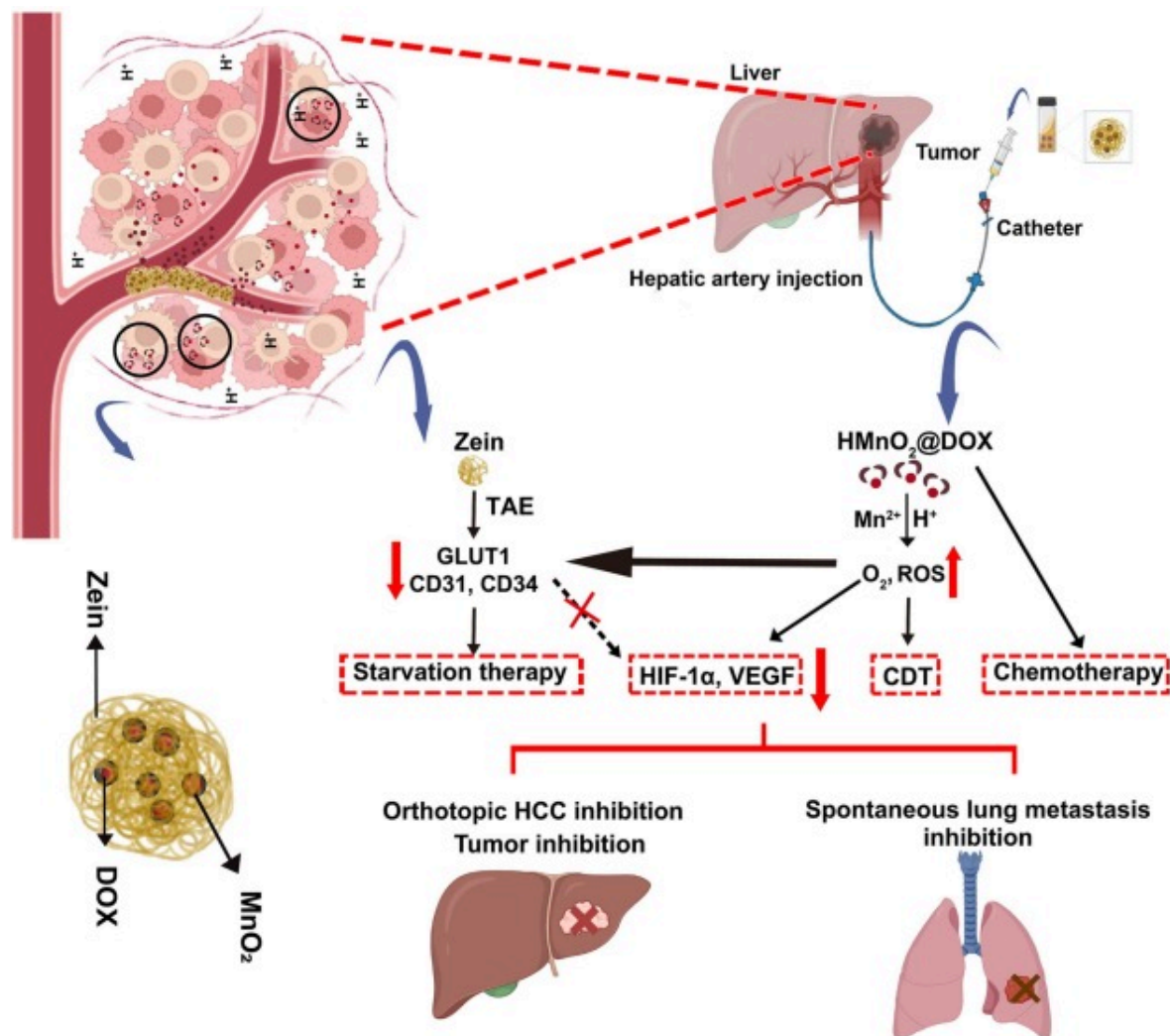
Early clinical exploration of zein dates back to its use in an alcoholic solution (Ethibloc®) for the percutaneous embolization of aneurysmal bone cysts, demonstrating efficacy as a standalone sclerosant and embolic agent [92]. Recent research has focused on refining zein formulations to improve handling, radiopacity, and therapeutic functionality. For instance, a re-designed zein-based fluid agent was evaluated in a porcine renal embolization model. The formulation, whether used pure or mixed with iodized oil or ethanol, achieved effective distal vascular occlusion from segmental arteries down to glomerular capillaries, demonstrating its potential for deep penetration and complete vascular blockade [91].

Further development led to the novel zein-based non-adhesive precipitating liquid embolic HEIE1_2017, which combines zein, ethanol, and iodized oil for radiopacity. Preclinical studies in pig kidneys showed that this agent provided excellent visibility under X-ray and CT imaging and achieved homogeneous occlusion from large segmental arteries down to the microvasculature. When manually modified for adjusted viscosity, it could penetrate even deeper into peritubular capillaries, offering a tunable occlusion profile [93]. Compared to ethylene vinyl alcohol copolymer (EVOH)-based liquid embolics (Onyx®), zein-based formulations caused fewer imaging artifacts and showed promising mechanical stability.

Beyond simple occlusion, zein has been engineered as a drug-eluting platform for chemoembolization. A significant advancement is the development of a zein-based in situ gelling system for sustained release of sclerosing or chemotherapeutic agents. For example, pingyangmycin (PYM)-loaded zein/zein-SAIB (sucrose acetate isobutyrate) in situ gels were developed for the treatment of venous malformations. This formulation

gelled upon injection, providing extended local drug release over several days, which enhanced the sclerosing effect while potentially reducing systemic side effects [94,95].

Most recently, zein has been leveraged to address a major limitation of conventional TACE: the aggravation of a hypoxic and pro-metastatic tumor microenvironment. An innovative composite embolic agent was developed by encapsulating doxorubicin (DOX)-loaded mesoporous hollow manganese dioxide (HMnO₂) nanoparticles within a zein matrix (Zein-HMnO₂@DOX) (Fig. 8) [96]. This system exploits multiple responsive mechanisms: zein gelation upon contact with blood provides immediate mechanical embolization, while the acidic tumor microenvironment triggers the decomposition of HMnO₂. This decomposition not only releases DOX but also generates oxygen (alleviating hypoxia) and produces reactive oxygen species (ROS) via a Fenton-like reaction. In rabbit VX2 liver tumor models, this agent significantly outperformed conventional lipiodol TACE, inhibiting primary tumor growth and, crucially, downregulating hypoxia-inducible factor (HIF-1 α) and vascular endothelial growth factor (VEGF) to suppress lung metastasis [96]. This approach exemplifies the shift toward “smart” embolics that actively remodel the tumor microenvironment.



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Fig. 8. Zein composite embolic agents combining TACE therapy and CDT therapy. (Reproduced with permission from ref. [96]. Copyright 2024, Elsevier published under a Creative Commons license).

Zein's unique property of precipitating upon contact with aqueous environments allows it forming stable occlusions in blood vessels, making it an effective liquid embolic agent. This property, combined with its biodegradability and compatibility with hydrophobic drugs, enables zein to be formulated as a versatile platform for both occlusion and drug delivery. The recent development of zein-based composite embolics that respond to the tumor microenvironment (pH-sensitive release of doxorubicin and oxygen generation) is an example, how material design can be directed to overcome the limitations of conventional TACE, such as hypoxia-induced metastasis. Thus, zein represents a promising material for the next generation of 'smart' embolic agents that actively modulate the tumor microenvironment.

The key advantages of zein include its natural origin, biodegradability, capacity for deep vascular penetration (as a liquid embolic), and compatibility with hydrophobic drugs. Its precipitation mechanism allows for controlled occlusion. However, challenges remain, including the need for precise formulation to control viscosity and precipitation rate, potential inflammatory responses to ethanol solvents, and a relative lack of large-scale, long-term clinical data compared to established polymers like PVA or gelatin. Further clinical translation will require standardized manufacturing and comprehensive biocompatibility studies.

3.7. Alginate

Alginate, a naturally derived anionic polysaccharide extracted from brown seaweed, has emerged as a highly versatile and extensively studied biopolymer for vascular embolization due to its biocompatibility, biodegradability, and mild gelation behavior in the presence of divalent cations such as calcium (Ca^{2+}) [97]. Its ability to form hydrogels under physiological conditions allows it to be delivered as an injectable liquid that solidifies in situ, enabling deep and conformal vessel occlusion (Fig. 9).

emulsification-gelation or electrospray techniques, demonstrating effective renal and hepatic arterial embolization in animal models [100]. These microspheres can be loaded with chemotherapeutic agents like doxorubicin (DOX) or norcantharidin, providing sustained release profiles that enhance localized therapy while minimizing systemic exposure [101,102]. Furthermore, alginate serves as an excellent matrix for more sophisticated, stimulus-responsive systems. For instance, alginate microspheres have been used to encapsulate temperature-sensitive liposomes (TSL) loaded with DOX and MRI contrast agents. This design allows for MR-guided embolization and on-demand drug release triggered by mild hyperthermia, representing a significant step toward image-guided, spatially controlled chemoembolization [103].

A major limitation of many polymeric embolics is the lack of direct imaging capability. This has been addressed by incorporating radiopaque materials. Tantalum nanoparticles have been embedded into alginate microspheres (Ta@CaAlg), providing excellent and persistent X-ray visibility for real-time procedural guidance and long-term post-embolization monitoring [101]. Similarly, liquid metal (for example, gallium and indium) droplets or gallium nanorods have been incorporated into alginate hydrogels or microspheres, conferring radiopacity for CT tracking and enabling additional modalities like photothermal therapy [104,105].

For embolization in traumatic hemorrhage, alginate-calcium microspheres have been successfully loaded with thrombin (TACMs). These microspheres promote rapid and robust thrombus formation at the target site, offering a potent hemostatic embolic agent for treating solid organ injury [106,107].

Alginate's structure allows it to act as a platform for combination therapies. Microspheres encapsulating molybdenum disulfide (MoS₂) nanosheets serve as dual microwave sensitizing and embolic agents, enabling enhanced microwave ablation of large tumors following transarterial embolization [108]. Other composites combine alginate with silk fibroin to improve mechanical properties and drug loading [109], or with poloxamer 407

to create temperature-sensitive liquid embolics that gel at body temperature for deep vessel penetration [110].

Innovative approaches aim to counteract the adverse tumor microenvironment (TME) post-embolization. Calcium carbonate (CaCO_3)-encapsulated alginate microspheres have been developed to neutralize the acidic TME of hepatocellular carcinoma following transarterial embolization (TAE). This pH modulation was shown to reverse immunosuppressive conditions, enhance therapeutic efficacy, and improve survival in preclinical models, highlighting a strategic move toward immunomodulatory embolization [111].

The use of pH-responsive components like CaCO_3 to neutralize tumor acidity represents a method to remodel the tumor immune microenvironment (TIME). Post-embolization hypoxia and acidosis are potent factors leading to immunosuppression, increased expression of immune molecules, and recruitment of regulatory T cells. By mitigating acidosis, these 'smart' microspheres can reverse this immunosuppressive shift, potentially enhancing the infiltration and function of cytotoxic T lymphocytes and natural killer cells. This allows alginate-based embolics to be considered not only as occlusive agents but also as active TIME modulators, opening the way for combination treatment regimens with immunoinhibitors to improve long-term control of hepatocellular carcinoma.

Alginate's gentle gelation in the presence of calcium ions allows the formation of soft, conformable hydrogels that can fill vessels completely and adapt to their shape, leading to effective and durable embolization. The ability to incorporate radiopaque materials (e.g., tantalum) or therapeutic agents (e.g., chemotherapeutics, thrombin) into alginate microspheres demonstrates its versatility as a platform for imageable and drug-eluting embolics. Moreover, the biodegradability of alginate can be controlled by adjusting the cross-linking density and composition, enabling the design of embolics with tailored persistence. The development of alginate microspheres that neutralize the acidic tumor microenvironment further illustrates how material properties can be engineered to not only occlude vessels but also modify the tumor biology for enhanced therapeutic effect.

Despite of its potential, the clinical translation of alginate embolics faces several challenges. Precise control over gelation kinetics and rheology is critical to prevent premature catheter occlusion or distal migration. The long-term degradation profile and fate of alginate in vivo require further standardization. While animal studies are abundant, large-scale randomized clinical trials comparing alginate-based agents to established products (PVA microspheres and others) are needed to fully establish their clinical efficacy and safety profile. Nonetheless, alginate's versatility, biocompatibility, and capacity for sophisticated engineering position it as a foundational polymer for the next generation of multifunctional, imageable, and therapeutic embolic agents.

Thus, the most promising for the development of new embolizers of hepatocellular carcinoma vessels are microspheres based on biodegradable polymers, in particular gelatin, hyaluronic acid, alginate and starch. In turn, the range of applications of polymer compositions is wide and is not limited to embolization of vessels feeding hypervascularized tumors (such as hepatocellular carcinoma). Today, embolization with polymer compositions is of interest for preoperative embolization of vessels feeding hypervascularized benign formations of the head, neck and mediastinum, in order to reduce blood loss during surgical resection, as an organ-preserving treatment for uterine fibroids and as an alternative to surgical treatment of prostate hyperplasia and internal hemorrhoids.

To systematically compare the key properties and applications of the polymer matrices discussed in this section, [Table 3](#) provides an overview of their synthesis methods, biodegradability, drug-loading mechanisms, release kinetics, elasticity, and primary clinical indications. This comparative analysis underscores how material characteristics dictate functional performance and clinical utility, guiding the rational selection of microsphere compositions for specific embolization scenarios.

Table 3. Comparative overview of key polymer matrices used in embolic microspheres.

Polymer matrix	Synthesis methods	Biodegradability	Drug-loading mechanism	Release kinetics	Elasticity/deformal
Polyvinyl Alcohol (PVA)	Emulsion cross-linking, spray drying	Non-degradable or slow degradation (depends on cross-linking)	Ionic binding (sulfonate groups), absorption	Sustained release (hours–weeks), depends on cross-linking density	High elasticity, compressible
Chitosan	Emulsion cross-linking, membrane emulsification	Biodegradable (enzymatic)	Ionic interaction (amine groups), surface adsorption	pH-dependent, typically sustained	Moderate elasticity, be tuned by cross-linking
Gelatin	Emulsion cross-linking	Biodegradable (enzymatic)	Physical entrapment, electrostatic interaction	Sustained (days–weeks), tunable via cross-linking	Soft, deformable
Hyaluronic acid (HA)	Microfluidics, emulsion cross-linking	Biodegradable (hyaluronidase)	Covalent conjugation, physical entrapment	Sustained, degradation-controlled	Excellent elasticity, rapid recovery after compression
Starch	Emulsion cross-linking	Rapidly biodegradable (α -amylase)	Absorption, physical entrapment	Fast release, short-term occlusion	Limited elasticity

Polymer matrix	Synthesis methods	Biodegradability	Drug-loading mechanism	Release kinetics	Elasticity/deformal
Zein	Precipitation from ethanol solution, emulsification	Biodegradable (enzymatic)	Hydrophobic entrapment, covalent modification	Sustained, triggered by pH/environment	Tunable (from liquid embolic to solid microspheres)
Alginate	Ionic gelation (Ca^{2+}), emulsification-gelation	Biodegradable (ion exchange, slow hydrolysis)	Ionic/covalent binding, physical entrapment	Sustained, ion-exchange-controlled	Soft hydrogel, conformable

4. Embolization in patients with paragangliomas

Paragangliomas are benign tumors of neuroectodermal origin, originating from the paraganglionic cells of the adventitia of blood vessels or associated nerves [112].

Paragangliomas can occur in various locations, including the abdomen, pelvis, chest, head, neck [113], and mediastinum. Invasion or close connection of paragangliomas with blood vessels and lower cranial nerves, jugular bulb, facial nerves, internal carotid artery, their high vascularization, and poorly defined margins of paragangliomas often become insurmountable obstacles to their radical removal or a reason for refusing surgery [114].

Preoperative vascular embolization is an effective strategy for ensuring safe surgical intervention, as it reduces intraoperative blood loss, operative time, simplifies surgical intervention, improves visualization, which allows for radical excision without serious complications [115]. Among the most difficult for surgical intervention are

paragangliomas of the head, neck and mediastinum. As embolizing particles for preoperative embolization, mainly PVA-based microspheres and liquid compositions based on ethylene-vinyl alcohol copolymer are used ([Table 4](#)).

Table 4. Compositions for embolization of paraganglioma vessels.

Registered name	Manufacturer	Composition/Parameters
Contour SE	Boston Scientific, Natick, MA, USA	Polyvinyl alcohol. Diameter from 45 to 150µm to 1000–1180µm.
Onyx® Liquid Embolic System	ev3 Inc. Plymouth, MN, USA	Components: - embolizing component: ethylene-vinyl alcohol copolymer; - solvent – DMSO. Contrast component for radiography – micronized tantalum.

Sanna M. et al. performed intra-arterial embolization using PVA particles in six patients (two women and four men) with progressive paraganglioma of the temporal bone and skull base 48h before surgery. Embolization reduced the duration of surgery and the volume of operative blood loss [[116](#)]. In the work [[117](#)], a 58-year-old woman with mediastinal paraganglioma receiving blood supply from the coronary and bronchial arteries underwent embolization of mediastinal branches using a combination of coils and polyvinyl alcohol microspheres Contour SE (Boston Scientific, Natick, Ma, USA) measuring 355–500µm, followed by complete surgical resection *en bloc*. According to angiography data, embolization led to almost complete cessation of blood flow in the coronary and bronchial branches of the mediastinum. Three-month follow-up of the patient after embolization and resection showed no symptoms of mediastinal paraganglioma (nausea, confusion), no signs of edema, and no abnormalities in vital signs (serum sodium levels within normal limits). In a study [[118](#)] performed by the

Department of interventional radiology (Guangzhou First People's Hospital, Guangzhou, China), according to intraoperative and postoperative information on patients (130 patients with embolization and 39 patients without embolization), a decrease in the average size of carotid paraganglioma, average operative time, and total number of complications was found in patients with preoperative embolization ([Table 5](#)).

Table 5. Indicators for patients with and without preoperative embolization [[118](#)].

Parameter	Without embolization	With embolization
Average size of paraganglioma, mm	43.49	35.44
Average operation time, min	195.48	205.64
Intraoperative blood loss, cm ³	215.15	251.41
Overall percentage of complications, %	26	36

A systematic review by the collaborators from Faculty of Medicine, Max Planck University Center, São Paulo, Brazil, involving 106 patients with carotid paraganglioma aged 18 to 79years, found total devascularization in 97% of cases with a low complication rate after embolization (1%) [[119](#)]. Preoperative embolization was performed using the liquid composition Onyx, consisting of a copolymer of ethylene with vinyl alcohol, dimethyl sulfoxide as a solvent, and micronized tantalum as a contrast agent for radiography.

Thus, preoperative arterial embolization made surgical resection of paragangliomas safer for patients and reduced the risk of postoperative minor and major complications. It should also be noted that, compared with coil embolization, microsphere embolization allows complete filling of the vascular bed. Complete filling of the vessel is of great importance, since the unfilled part of the paraganglioma vascular network with insufficient penetration of the embolic agent into small vessels increases the risk of intraoperative complications and leads to the development of collateral blood flow and relapses [[120](#)].

The choice of embolic agent in paraganglioma embolization is guided by the need for deep, permanent occlusion of hypervascular tumor formations without damaging neighboring critical structures. Spherical, non-adhesive microspheres (e.g., Contour SE) offer predictable distal penetration due to their uniform size and deformability, enabling complete devascularization while minimizing the risk of inadvertent embolization of cranial nerve vasculature. In contrast, liquid embolics like Onyx (ethylene-vinyl alcohol copolymer) can achieve deeper penetration but carry a higher risk of uncontrolled distal migration and non-target ischemia. The material's radiopacity, provided by tantalum in Onyx, enhances procedural safety by allowing real-time fluoroscopic monitoring, a property, which is lacking in many polymeric microspheres. These material characteristics directly influence surgical outcomes: studies show that spherical microspheres reduce intraoperative blood loss more effectively than coils or irregular particles, owing to their ability to occlude at the capillary level [121].

However, in the field of paraganglioma treatment, additional studies are needed to take into account the influence of individual tumor characteristics, size and anatomy of tumor-feeding vessels on surgical efficacy and safety [122]. In addition, the influence of the surgeon's skill level, patient characteristics and institutional practice on intraoperative time and estimated blood loss during surgery should be considered. Limitations of the presented studies that affect the consistency and accuracy of the data also include the assessment of complications only after surgery, rather than after embolization and after surgery, and the variability in blood loss estimates reported by surgeons.

5. Embolization in patients with uterine fibroids

Uterine fibroids are the most common benign uterine tumors affecting women of reproductive age [123]. In 25–30% of patients, uterine fibroids are associated with symptoms such as menorrhagia, pelvic pain, bowel and urinary symptoms, and infertility [124]. Medical treatment options for uterine fibroids are limited, and available drugs (progestogens, oral contraceptives, gonadotropin-releasing hormone agonists and antagonists) are used for short-term treatment of bleeding in patients with fibroids [125].

Traditional surgical treatments involve either complete removal of the uterus (hysterectomy) or removal of the fibroids (myomectomy). Uterine artery embolization (UAE) is a non-surgical, minimally invasive method that allows preserving the uterus [126].

Patients who underwent UAE have no intraoperative bleeding, a lower rate of hemoglobin decline, shorter operative, postoperative and overall hospitalization, and a lower rate of postoperative recurrence compared to patients who underwent laparoscopic myomectomy (LM) [127] (Table 5). UAE treatment is also less invasive, has a faster recovery, and has a lower recurrence rate. However, the hospitalization cost for UAE is higher compared to LM (Table 6).

Table 6. Surgical and hospital clinical parameters in groups of patients with UAE and LM [127].

Indicator	UAE group (n=66)	LM group (n=66)
Blood loss (mL)	0	140.91
Operation duration (minutes)	58.71±16.49	79.82±36.79
Rate of hemoglobin decline [(preoperative hemoglobin value–postoperative hemoglobin value)/preoperative hemoglobin value]×100%	7.24±8.67	13.43±6.04
Postoperative hospital stay, days	2.67±1.27	4.95±0.64
Hospital stay (days)	4.50±2.12	7.26±0.95
Time to resume work or social activities after surgery (days)	16.42±11.48	27.70±9.22
Total number of relapses, over 2years	11 (16.7%)	21 (31.8%)
Hospitalization cost (yuan)	24,400.06±6095.98	11,815.06±1706.71

Compared with hysterectomy, UAE was also associated with shorter hospital stay, less blood transfusion, fewer cases of pelvic floor prolapse, bowel obstruction, and re-interventions [128].

However, UAE may result in post-embolization syndrome, which is characterized by severe pain 6–8 h after UAE and by inflammation [129]. To minimize the risks of post-embolization syndrome, polyvinyl alcohol (PVA) particles, tris-acryl gelatin microspheres (TAGM), polyethylene glycol microspheres, and polymethyl methacrylate microspheres coated with poly[bis(trifluoroethoxy) phosphazene] are predominantly used as embolic agents (Table 7).

Table 7. Microparticles for uterine artery embolization.

Registered name	Manufacturer	Composition/Parameters
Bearing nsPVA	Merit Medical	Biocompatible hydrophilic non-absorbable particles of irregular shape. Made from polyvinyl alcohol. Size: 45–150, 150–250, 250–355, 355–500, 500–710, 710–1000, 1000–1180 μm .
Embospheres	Biosphere Medical, Rockland, MA, USA	Trisacryl gelatin (microspheres of cross-linked trisacryl coated with gelatin). Size: 40–120, 100–300, 500–700, 700–900, 900–1200 μm .
HydroPearl™	Terumo Interventional Systems, Tokyo, Japan	Compressible microspheres for embolization. Polyethylene glycol in phosphate-buffered saline. Size: 5 ± 25 , 200 ± 75 , 400 ± 75 , 600 ± 75 , 800 ± 75 , $1100 \pm 75 \mu\text{m}$.
Embozene Color-Advanced Microspheres	CeloNova BioSciences, Newnan, Georgia, USA	Polymethyl methacrylate microspheres coated with polyphosphazene (Polyzene®-F,

Registered name	Manufacturer	Composition/Parameters
		poly[bis(trifluoroethoxy) phosphazene]) Size: 40, 100, 250, 400, 500, 700, 900 μ m.

The authors of [130] conducted a study involving 54 women with uterine fibroids to evaluate the short-term therapeutic effects, pain, and inflammatory response after UAE using PVA microparticles (Bearing, Merit Medical, 355–500 μ m) and tris-acrylgelatin microparticles (Embosphere, Biosphere Medical, 500–700 μ m). Embolization was performed until the blood flow in the ascending uterine artery was completely stopped. The authors found that there was no statistically significant difference in inflammatory markers such as the number of leukocytes, the percentage of neutrophils and lymphocytes, and the content of C-reactive protein in the blood of patients before UAE and 24h after UAE. However, the change in the neutrophil and lymphocyte ratio was higher in the group with PVA particles compared to tris-acrylgelatin particles, and the inflammatory response was more pronounced. This is due to the fact that the authors used PVA particles with a wide granulometric size range and irregular shape, which leads to their aggregation before reaching the vascular network of the fibroid and causes embolic effects in the uterus. Thus, TAGM particles are more promising agents for UAE than PVA particles.

Along with tris-acrylgelatin microspheres, it was shown that embolization with HydroPearl polyethyleneglycol microspheres (Terumo Interventional System, Tokyo, Japan) is also a safe and effective treatment option for symptomatic uterine fibroids. HydroPearl microspheres are easily compressed and maintain a spherical shape after passing through a microcatheter, which allows for predictable embolization. In [131], 800- μ m HydroPearl microparticles suspended in saline were used for uterine artery embolization in 37 patients with uterine fibroids. The microparticles were injected until complete blood flow congestion was achieved for at least 3–5 cardiac cycles. As a result, the authors observed a decrease in the diameters (from 62.3 to 52.7mm) and volumes

(from 183.1 to 154.6cm³) of uterine fibroids. The observed clinical response rates for menorrhagia and mass symptoms were similar to those of previously studied tris-acrylgelatin-based embolic hydrogel microspheres (Embospheres) and polyphosphazene-coated polymethyl methacrylate microspheres (Embozoene) [132].

It should be also noted that post-embolization outcomes for large and small particles associated with menorrhagia, pelvic pain and volumetric symptoms are similar. In particular, in the study [133] involving 76 patients, the absence of a statistically significant relationship was shown between the particle size of Embosphere microspheres (particles 300–500µm and 500–700µm) and (500 and 700µm) and post-embolization outcomes.

However, the clinical studies reviewed have limitations that could have affected the results. The limitations are related to the lack of specific inclusion criteria for patients and different uterine fibroid morphology (size, location), limited sample size due to patients who dropped out of observation and did not complete the questionnaire.

Material properties critically determine the inflammatory response and clinical success of uterine artery embolization (UAE). Smooth, spherical microspheres (e.g., Embospheres, HydroPearl) cause less endothelial trauma and platelet activation than irregular PVA particles, resulting in lower levels of post-procedural pain and inflammatory markers (C-reactive protein, neutrophil-to-lymphocyte ratio). Their compressibility allows uniform delivery through microcatheters without aggregation, leading to predictable occlusion of perfibroid vessels and sustained ischemia. In contrast, irregular PVA particles tend to clump and occlude proximally, increasing the risk of non-target uterine ischemia and post-embolization syndrome. Furthermore, the biodegradability of certain materials (gelatin, starch) may influence long-term outcomes: while non-degradable particles provide permanent occlusion, degradable agents may allow revascularization over time, potentially affecting recurrence rates.

6. Prostatic artery embolization

The prevalence of benign prostatic hyperplasia (BPH) is 60% of men aged 60 years and increases with age [134]. Almost one in three men with BPH has developed lower urinary tract symptoms, which are associated with worsening in quality of life [135]. Drug therapy with α -blockers and 5 α -reductase inhibitors reduces the risk of clinical progression, but one in ten patients will still experience recurrent urinary tract infection [136].

Traditional surgical approaches (transurethral resection of the prostate, laparoscopic adenomectomy) improve symptoms in patients with BPH. However, elderly patients have an increased risk of complications after surgery associated with urinary incontinence, retrograde ejaculation, hematuria, urethral stricture, and bladder neck sclerosis [137,138].

Prostatic artery embolization (PAE), which involves occlusion of the prostate blood supply with subsequent necrosis, reduction of bladder outlet obstruction and prostate shrinkage, is a new method for the relief of lower urinary tract symptoms in BPH [139,140]. The advantage of PAE over traditional surgical approaches is the absence of serious complications and the fact that it is performed under local anesthesia, allowing for a shorter hospital stay [141].

PAE is also effective in elderly patients. In the work [142], the efficacy and safety of PAE was assessed in ten patients aged 74–87 years with resistance to drug therapy, an average prostate volume of 66.4 cm³, and an average maximum urine flow rate $Q_{\max}$ of 8.7 mL/s. Embolization was performed through a microcatheter into the prostatic artery using Embosphere tris-acrylgelatin microspheres of 300–500 μ m in size mixed with a contrast agent and saline. Pelvic angiography was used to identify the prostate arteries [142].

As a result, clinical success of embolization was achieved for all patients, consisting in a decrease in the mean maximum urine flow rate $Q_{\max}$ from 8.7 to 12.6 mL/s and the mean prostate volume from 66.4 to 49.4 cm³. From a safety point of view, no possible serious complications were observed after 24 months of follow-up, including bladder ischemia and persistent perineal pain.

PAE can also be performed in older patients (>87years). For example, in the study [143], successful PAE was performed in a 100-year-old man with severe lower urinary tract symptoms (refractory infection, recurrent gross hematuria and nocturia) and an enlarged prostate. Embolization was performed by injecting Embosphere microspheres mixed with 10mL of iodinated contrast medium through a catheter into the left prostatic artery. After embolization, the patient's quality of life improved significantly. Magnetic resonance imaging performed 24months later showed a 35% decrease in prostate volume [143].

However, in the studies discussed above, only tri-sacryl-gelatin microspheres were used for PAE. In contrast, in studies [[144], [145], [146]], the safety and efficacy of not only Embospheres tri-sacryl-gelatin microspheres (sizes 100–300, 300–500 μm), but also Embozene microspheres (250, 400 μm) and polyvinyl alcohol-based microspheres (100–300 μm , Merit Medical Systems, Inc.) were assessed. As a result, an improvement in the efficacy criteria (Q_{max} , prostate volume, prostate-specific antigen, presence of adenomatous-dominant benign prostatic hyperplasia) and safety was determined in all groups at 6 and 12months after embolization, as well as the absence of statistically significant differences in the criteria for the three examined groups. A study of the effect of particle size on clinical efficacy showed favorable clinical results for smaller particles compared to larger ones ($\geq 300\mu\text{m}$). This is likely due to the fact that smaller microspheres more effectively penetrate the prostate vasculature, leading to tissue necrosis and a decrease in prostate size. This assumption was confirmed by Miller S.R. et al. [147]. It was determined that trisacryl-gelatin microsphere particles of 100–300 μm in size have the maximum distal penetration into the prostatic artery in a microvascular model in vitro, and trisacryl-gelatin microsphere particles of 300–500 μm in size have the least distal penetration.

Liquid embolization is used to eliminate gaps between polymer microspheres after injection. The combined use of polyvinyl alcohol microspheres with the thermosensitive liquid embolic agent TempSLE based on poly(N-isopropylacrylamide) and iohexol dissolved in saline promotes the formation of cast within arteries, increases the rate of necrosis of the glandular tissue of the prostate gland and leads to a decrease in the

volume of the prostate. At temperatures below 30°C, poly(N-isopropylacrylamide) is in a liquid state and forms a gel at human body temperatures, which allows it to fill the gaps between the microspheres in the artery. In addition, the combined use of microspheres and TempSLE is safe for patients and does not cause ectopic embolism, ischemia of the penis, bladder wall and rectum [148].

The clinical efficacy of PAE is linked with microsphere size and material composition. Smaller microspheres (100–300µm) penetrate deeper into the prostatic microvasculature, leading to more extensive glandular necrosis and greater volume reduction, which are key factors of symptom improvement. However, excessive distal penetration also increases the risk of non-target ischemia (bladder wall, rectum). The spherical shape and smooth surface of trisacryl-gelatin microspheres (Embospheres) reduce aggregation and catheter clogging, ensuring consistent delivery. In contrast, irregular PVA particles may occlude variably, contributing to inconsistent clinical responses. The emergence of liquid embolics (e.g., TempSLE) that gel in situ addresses the limitation of particle settling and incomplete filling, potentially improving occlusion durability. These material properties directly influence functional outcomes such as International Prostate Symptom Score (IPSS) reduction, peak urinary flow rate (Q_{max}) improvement, and the incidence of complications like retrograde ejaculation.

However, the limitations of the conducted clinical trials are the use of different volumes of embolic injection compositions, the assessment of medium-term but not long-term (more than 2years) outcomes, the lack of comparison with the outcomes of other methods of minimally invasive therapy of benign prostatic hyperplasia, such as holmium laser enucleation of the prostate (HoLEP) and water vapor therapy (Rezüm).

In addition, the success of prostate artery embolization often depends on the characteristics and architecture of the vascular bed of a particular patient, the experience and technique of the X-ray endovascular surgeon. The latter emphasizes that the results of studies by different authors sometimes differ significantly in terms of both the effectiveness of treatment and the technical feasibility of performing the procedure itself.

7. Embolization of hemorrhoidal arteries

Hemorrhoidal disease is the most common anorectal pathology [149]. Although approximately 10% of patients eventually undergo surgery, various contraindications to surgery and painful and tedious postoperative care have prompted new minimally invasive procedures [150]. Embolization of hemorrhoidal artery blood flow is a promising alternative to surgical methods of hemorrhoid treatment, such as Milligan-Morgan hemorrhoidectomy, hemorrhoidal ligation, and Doppler-guided ligation of the superior rectal artery branches [151]. Embolization, in contrast to surgery, is a painless treatment method, allows preserving anal tone without direct anorectal trauma and normal anal anatomy, provides rapid recovery, and requires minimal local wound care in an outpatient setting [151]. Microspheres are used for embolization of hemorrhoidal arteries in patients with recurrent bleeding, anemia, and chronic hematochezia. Microspheres, unlike microspirals, can close the hemorrhoidal plexus more distally and prevent anastomoses of the superior and middle rectal artery in patients with symptomatic hemorrhoidal disease [152].

Embozene microspheres based on polymethyl methacrylate coated with polyphosphazene and tris-acrylgelatin microspheres are used for embolization of hemorrhoidal arteries, and the greatest reduction in hemorrhoidal bleeding rates is observed for larger microspheres (diameter 900–1200 μm). In the work [152], embolization with Embozene microspheres with an average diameter of 1100 μm , performed for 10 patients, led to a decrease in the hemorrhoid severity score after 12 months of observation. However, one patient experienced a serious complication – perforation of the rectosigmoid muscle, which led to the need for surgical treatment (Hartmann operation). Thus, further studies with a larger number of patients and observation for more than two years are required to confirm the safety of embolization with Embozene microspheres.

In turn, a study involving 42 patients with recurrent hemorrhoids with an average age of 45 years showed that the use of Embosphere tris-acrylgelatin microspheres with a size of

900–1200µm has a lower incidence of postoperative pain compared with Embosphere microspheres with a size of 500–700µm and 700–900µm [153] (Table 8).

Table 8. Percentage of postoperative pain in patients depending on the size of embolizing microspheres [153].

Size of Embos-phere microspheres, µm	Percentage of occurrence of postoperative pain, %
500–700	100
700–900	64.3
900–1200	14.3

The only complications after microsphere embolization were small (less than 5 mm) superficial ulcers in the area of the rectosigmoid junction and small fibrous ulcers, which, according to rectoscopy data, healed 6 months after treatment with mesalazine (Salofalk).

The success of hemorrhoidal artery embolization depends on precise occlusion of terminal arterioles without causing ischemic damage to the rectal mucosa. Larger microspheres (900–1200µm) occlude more proximally, reducing the risk of distal mucosal ischemia and ulceration, but can leave collateral channels patent. Smaller particles (500–700µm) achieve deeper occlusion but increase the possibility of post-procedural pain and tissue necrosis. The spherical geometry and smooth surface of modern microspheres (Embospheres, Embozene) prevent aggregation and enable controlled delivery, whereas irregular particles may cause unpredictable embolization. The non-biodegradable nature of most microspheres provides permanent occlusion, which may be advantageous for long-term symptom control but precludes repeat intervention if needed. These material considerations directly correlate with clinical endpoints such as bleeding recurrence, pain scores, and healing of mucosal lesions.

Summarizing, rectal artery embolization is a promising treatment for internal hemorrhoids in patients with recurrent hemorrhoids. Despite the significant suffering of hemorrhoid patients and the economic burden, research in the field of embolization treatment of hemorrhoids is limited. Further comparative studies on bleeding, postoperative pain, ischemic complications for embolization with microparticles synthesized from different polymers, and the effectiveness of microsphere embolization with other treatment methods are needed. Also, to fully evaluate the role of microsphere embolization in the treatment of hemorrhoids, future studies should focus on accurate patient selection and long-term clinical outcomes.

8. Critical considerations: Biocompatibility, stability and limitations

The clinical translation and long-term success of polymer microspheres in endovascular therapy are depend not only on their primary function, vascular occlusion and drug delivery, but also on addressing several interconnected critical challenges. These include ensuring biocompatibility and hemocompatibility, maintaining chemical and physical stability, and mitigating inherent limitations such as physical erosion and non-target effects. A systematic evaluation of these factors is essential for the rational design of safe and effective embolic agents.

8.1. Biocompatibility and hemocompatibility

Biocompatibility refers to the ability of a material to perform its intended function without eliciting any undesirable local or systemic effects in the host. For intravascular implants like embolic microspheres, this extends to specific hemocompatibility, a compatibility with blood. Upon injection, microspheres interact directly with blood components and the vascular endothelium, making strict hemocompatibility evaluation mandatory [[154,155](#)].

The implantation of microspheres triggers a cascade of biological responses [[156](#)]. An acute inflammatory reaction is typical, but it should resolve into a thin fibrous capsule for

stable integration. Excessive or chronic inflammation can lead to pain (post-embolization syndrome), vessel wall damage, or recanalization [157]. For example, the cationic nature of chitosan, while beneficial for hemostasis, can sometimes provoke heightened inflammatory responses if not carefully controlled through purification and cross-linking [158]. Similarly, residual aldehydes from glutaraldehyde cross-linking in gelatin or PVA microspheres can cause cytotoxicity and must be meticulously removed [159].

Hemocompatibility testing also involves evaluating thrombogenicity (potential to cause clots), platelet activation, complement activation, and hemolysis (red blood cell damage). Standardized ISO 10993 tests (cytotoxicity, sensitization, intracutaneous reactivity, systemic toxicity, implantation) form the basis of preclinical embolizing systems safety assessment. The degradation products of the polymer must also be evaluated. For instance, acidic by-products from the hydrolysis of poly-L-lactide can incite inflammation if not adequately buffered in vivo [160].

Poor biocompatibility can manifest as serious adverse events, including non-target ischemia, chronic pain, inflammatory pseudoaneurysms, or systemic inflammatory response syndrome. Therefore, the selection of polymer grade, purification methods, cross-linking agents, and sterilization processes (gamma irradiation, ethylene oxide, etc.) are critical design parameters that directly influence biological safety.

8.2. Microsphere stability: Chemical, physical, and functional

The stability of embolic microspheres is a multi-factored characteristic including their chemical integrity, physical form, and functional capacity over time, from manufacturing through storage and to deployment in vivo.

Chemical stability involves the resistance of the polymer matrix to degradation processes such as hydrolysis, oxidation or enzymatic breakdown during the storage. Chemical instability can weaken microspheres before use, leading to premature degradation and altered release kinetics. The molecular weight and crystallinity of the polymer are key

determinants. For example, high molecular weight, highly cross-linked PVA is more chemically stable than loosely cross-linked gelatin [161].

Physical stability refers to the maintenance of critical physical attributes, such as size distribution, shape, surface morphology, and mechanical strength (elasticity, compressive modulus). Physical instability can lead to aggregation of microspheres in suspension, causing catheter clogging and unpredictable embolization endpoints. Early irregular PVA particles were notorious for aggregation and fragmentation, a problem largely addressed by the development of smooth, spherical microspheres (for example Embospheres, Embozene) [16,162]. The method of sterilization can also impact physical stability, potentially making microsphere more fragile.

In addition to the above, for drug-eluting beads (DEB) or radioembolization agents, functional stability means the retained ability to carry and control the release of the active agent. This includes preventing drug leaching during storage and maintaining the radionuclide firmly bound to the matrix. The ionic binding of doxorubicin to sulfonated PVA in DC Beads or CalliSpheres is designed to be stable in storage but responsive to the ionic environment in vivo [163,164].

Ultimately, guaranteeing stability is a combined regulatory and scientific task. It demands comprehensive sterilization validation and shelf-life studies under various environmental conditions (temperature, humidity) to develop proper storage specifications and determine expiration dates. This comprehensive approach to stability, ensuring chemical, physical and functional integrity, transforms a promising polymer formulation into a reliable, predictable, and safe clinical tool, able to perform exactly as intended inside the patient's vasculature.

8.3. Limitations: erosion, non-target embolization and visualization

Despite significant advancements in material science and clinical technique, the application of polymer microspheres in endovascular therapy continues to be influenced by several limitations. These challenges are physical erosion, non-target embolization,

and the lack of direct visualization, which directly impact the safety profile, efficacy, and predictability of embolization procedures, actualizing continuing research and refining clinical protocols.

Once administered to the dynamic environment of the vasculature, microspheres are affected by hemodynamic forces. This can lead to physical erosion and mechanical fatigue, particularly in microspheres composed of polymers with low cross-linking density or insufficient inner bonding [72,165]. Erosion and fragmentation of the microsphere surface, lead to the set of adverse effects. Firstly, fragmentation can cause the premature and uncontrolled release of encapsulated chemotherapeutic agents or radionuclides. This disrupts the designed sustained-release pharmacokinetic profile, potentially leading to a toxic systemic burst and decreasing localized therapeutic effect [166]. Secondly, the smaller particles resulted from erosion process can travel far away the intended occlusion site. This distal embolization of capillary networks of non-target organs, such as the gallbladder, stomach, or lungs, is a known mechanism behind complications like cholecystitis, gastrointestinal ulceration, or pneumonitis [167]. Finally, the progressive reduction in occlusive volume due to material loss can result in partial recanalization of the embolized vessel. This revascularization pathway allows for tumor regrowth and represents a primary mode of treatment failure. This risk particularly increases when using very rapidly degradable embolics which occlusion duration is shorter than the time required for complete tumor necrosis [168].

Non-target embolization (NTE) remains a major problem, extending beyond the secondary effects of erosion. Primary NTE can occur due to a mismatch between microsphere size and the target vasculature, where particles smaller than the intended arterioles pass through the tumor bed into the venous circulation. Excessive injection pressure or the presence of arteriovenous shunts within hypervascular tumors can further intensify this effect. The consequences are severe and well-documented, including liver abscess formation, bile duct injury, and pulmonary embolism, which collectively represent a significant portion of procedure-related morbidity [167,169].

The mentioned above issues are compounded by lack of direct visualization for most conventional polymeric microspheres, which are radiolucent. This makes the embolization procedure partially “blind,” as interventional radiologists cannot directly visualize the distribution or density of the microspheres in real-time using standard X-ray fluoroscopy or computed tomography (CT). Medics must instead rely on surrogate angiographic endpoints, such as contrast stasis, which are subjective and may not correlate precisely with the completeness of tumor vessel occlusion or the extent of non-target spread [170,171]. This limitation is especially critical in transarterial radioembolization (TARE), where accurate post-procedural dosimetry is essential for evaluating treatment efficacy and safety. In response, a major focus of contemporary research is the development of imageable microspheres. Strategies include the incorporation of radiopaque markers like micronized tantalum or iodine into liquid embolics, and the design of microspheres visible under magnetic resonance imaging (MRI), aiming to transform embolization into a visually guided and quantitatively precise therapy [[172], [173], [174]].

In response to this critical need, significant research is dedicated to developing imageable microspheres. These next-generation agents integrate contrast-enhancing materials directly into their polymer matrix. Strategies include: (1) Incorporating radiopaque elements like micronized tantalum, iodine compounds, or bismuth nanoparticles for real-time X-ray/CT fluoroscopic tracking during administration; (2) Embedding superparamagnetic iron oxide (SPIO) or gadolinium chelates for Magnetic Resonance Imaging (MRI) visibility, allowing for superior soft-tissue contrast and post-procedural monitoring of distribution; and (3) Designing theranostic particles that combine therapeutic radioisotopes (Yttrium-90, Holmium-166) with gamma-emitting properties suitable for Single-Photon Emission Computed Tomography (SPECT), enabling post-treatment dosimetry. These advancements aim to transform embolization from a procedure guided by surrogate endpoints into a quantitatively precise, image-guided therapy, improving safety by minimizing non-target embolization and allowing personalized dosimetry.

Summarizing, the journey from a promising polymer formulation to a clinically successful embolic microsphere needs a deep and parallel focus on these mentioned above critical considerations. Future development must integrate advanced material science with rigorous biological testing to create next-generation agents that are not only effective occluders but also predictably stable, truly biocompatible, and visible, thereby maximizing therapeutic benefit while minimizing risk across all clinical applications discussed in this review.

9. Comparative analysis with alternative treatment methodologies

The clinical integration of microsphere embolization cannot be evaluated in isolation. Its value and appropriate application are best understood through direct comparison with the established therapeutic methods for each target condition. This technique does not represent a universal replacement but rather a strategic addition to the interventional arsenal, offering a number of benefits and limitations that must be estimated against surgical and alternative minimally invasive options.

In the management of hepatocellular carcinoma (HCC), microsphere-based transarterial chemoembolization (TACE) and radioembolization (TARE) occupy a middle ground [175]. They offer a more targeted and potent local effect than systemic therapies (tyrosine kinase inhibitors, immunotherapy) while being far less invasive than curative-intent surgical resection or transplantation. Compared to conventional lipiodol-based TACE, drug-eluting bead TACE (DEB-TACE) provides superior control over drug pharmacokinetics, leading to reduced systemic exposure and toxicity [176]. When contrasted with thermal ablation techniques (radiofrequency or microwave ablation), TACE and TARE can treat larger, multifocal, or ill-defined tumors, albeit potentially with a higher rate of local recurrence compared to complete ablation [177]. Thus, embolization role is often complementary, used for disease down-staging, preparation to transplantation, or treating patients with advanced disease or compromised liver function where surgery is contraindicated.

For paragangliomas, particularly those in complex anatomic locations, preoperative embolization with microspheres is compared to other embolic strategies [178]. Metallic coils provide rapid and secure proximal occlusion but often fail to penetrate the distal, hypervascular tumor nidus, potentially leaving a collateral network that complicates surgery. In contrast, liquid embolic (for example, ethylene-vinyl alcohol copolymer) can achieve deep penetration but carry a risk of uncontrolled distal migration and non-target embolization of cranial nerves or critical vessels. Spherical embolic microparticles offer a favorable compromise, enabling predictable, distal vessel-level occlusion with a lower risk of dangerous migration than liquids, thereby effectively reducing intraoperative blood loss and facilitating surgical intervention.

Within gynecology, uterine artery embolization (UAE) with microspheres presents a modern, uterus-preserving alternative [179]. Compared to the definitive but irreversible solution of hysterectomy, UAE preserves fertility potential and is associated with shorter hospital stays and faster recovery. When compared to laparoscopic or open myomectomy, UAE is significantly less invasive and avoids surgical scars and the risks of uterine rupture in future pregnancies, though it may be less effective for very large fibroids and also carries a risk of post-embolization syndrome. Compared to newer, incision-free energy-based therapies like magnetic resonance-guided focused ultrasound (MRgFUS) or radiofrequency ablation (RFA), UAE have longer experience of clinical application, better applicability to multiple and larger fibroids, and more reliable long-term outcome data.

The treatment of benign prostatic hyperplasia (BPH) has been revolutionized by minimally invasive options, among which prostatic artery embolization (PAE) occupies a specific niche [180]. The historical gold standard, transurethral resection of the prostate (TURP), provides the most rapid and substantial improvement in urinary symptoms but is associated with higher risks of bleeding, incontinence, retrograde ejaculation, and requires spinal or general anesthesia. Newer minimally invasive surgical therapies (MIST) like holmium laser enucleation (HoLEP) or convective water vapor therapy (Rezum) offer good efficacy with improved safety profiles over TURP. PAE differentiates itself by offering moderate but clinically meaningful symptom relief with an exceptional safety profile

regarding sexual function and urinary continence, positioning it as a compelling option for older patients with concomitant diseases or for those who prioritize the preservation of ejaculatory function.

In proctology, hemorrhoidal artery embolization positioned as a truly minimally invasive alternative for managing internal hemorrhoids [181]. It stands in contrast to excisional hemorrhoidectomy, which, while definitive, involves significant postoperative pain and a prolonged recovery period. Compared to office-based procedures like rubber band ligation, which are effective for low-grade hemorrhoids but with notable recurrence rates, embolization is more comprehensive in addressing vascular inflow. Doppler-guided hemorrhoidal artery ligation (HAL) shares a similar physiologic principle but is a surgical procedure. Embolization, performed percutaneously, offers the advantage of minimal post-procedural pain and a rapid return to normal activities, although its long-term durability and efficacy for advanced (Grade IV) hemorrhoids remain areas of ongoing investigation in comparison with traditional surgical approaches.

In summary, microsphere embolization does not displace existing treatments but strategically expands the therapeutic arsenal of methods available. Its core value proposition lies in a favorable risk-recovery profile, making it particularly valuable for patients who are suboptimal surgical candidates, those who desire organ preservation and individuals who prioritize minimally invasive methods of treatment and rapid functional recovery. Its development and implementation are connected with this comparative context, defining its role as a complementary and often patient-oriented choice in modern interventional medicine.

10. Conclusions

This review has synthesized the current landscape of polymer microsphere embolization across five clinical disciplines (oncology (HCC), vascular surgery (paragangliomas), gynecology (uterine fibroids), urology (BPH), and proctology (hemorrhoids)), revealing a critical interplay between material design and therapeutic outcome. Beyond a

compilation of applications, our analysis demonstrates that clinical efficacy is fundamentally governed by a core set of engineered material properties.

In the process of developing embolizing microspheres, the size, shape of the microspheres, molecular weight of the polymer, deformation of the microspheres, aggregation in the bloodstream, surface texture, absence of distal migration, which can lead to occlusion of large proximal vessels, pseudoembolization of the target vessel, are of great importance. Taking into account these factors, optimizing the properties of the microspheres will allow achieving controlled release of the required dose of a chemotherapeutic agent or radionuclide, reducing fluctuations in the concentration of the chemotherapeutic agent in the blood. In particular, microspheres of a larger diameter release less of the drug and more slowly than smaller microspheres. For polymers with a high molecular weight, the drug release profile is slower, since the microspheres are denser, and for low molecular weight polymers, it is faster. The use of a mixture of microspheres of different sizes will allow controlling the release of the chemotherapeutic agent.

In addition, knowledge of the dynamics of the release of a chemotherapeutic agent or radionuclide from microspheres is mandatory, since it allows selecting an individual treatment regimen for a specific patient. However, there is no unified method to determine desorption of a therapeutic agent. In our opinion, the most illustrative systems for studying the dynamics are model systems of arterial blood flow, simulating the processes of local embolization of arteries. This model is closest to real behavior in the human body. When developing methods for releasing a active agent from microspheres, it is necessary to take into account several parameters, such as desorption medium nature, the intensity of mixing, and the temperature.

Furthermore, biodegradability of the polymer matrix, which depends on the polymer crystallinity, molecular weight, and hydrophobicity, is of great importance for the development of microspheres. By changing these properties of the polymer matrix, it is possible to control the rate of biodegradation. This material characteristic critically

influences long-term safety and the possibility of re-treatment, creating a key difference between permanent occlusive agents and transitory therapeutic platforms.

However, in the works considered in current review and devoted to microspheres for embolization of hepatocellular carcinoma vessels, major attention is paid to apoptosis studies on laboratory animals, antitumor action on xenotransplant tumor models, in which human cancer cells are transplanted to mice and in a lesser extent the mentioned above factors are taken into account and studied. Therefore, in the future, for the development of effective microspheres, it is necessary to take into account the properties of polymer microspheres, the dynamics of the release of the chemotherapeutic agent or radionuclide into various physiological environments.

In turn, in the field of embolization of benign tumors and hemorrhoidal arteries using industrial brands of microspheres (Embospheres, Contour SE, Embozene Color-Advanced Microspheres, etc.), randomized clinical trials with a large sample size are required. Randomization with accidental distribution of participants will ensure greater objectivity of the study and minimize the influence of biases and errors on the results.

Looking forward, the convergence of material science and clinical insight must deepen. The systematic comparison of polymer matrices, as outlined in this review, underscores that there is no universal “best” material. Instead, the rational selection and design of microspheres must be guided by the specific pathophysiology of the target vessel and the desired therapeutic endpoint, whether permanent occlusion, sustained drug elution, or temporary vessel blockade. Future innovation should focus on “smart” designs that address current limitations, such as incorporating imaging capabilities for real-time guidance or modulating the post-embolization immune microenvironment. By prioritizing this structure-property-function paradigm, the next generation of embolic microspheres will achieve greater precision, efficacy, and personalization across all areas of endovascular therapy.

Abbreviations

A2AR Adenosine receptor controls cytokine production

BPH Benign prostatic hyperplasia

DEB-TACE Drug-eluting bead transarterial chemoembolization

DMSO Dimethyl sulfoxide

DNA Deoxyribonucleic acid

DSM Degradable starch microsphere

EMA European Medicines Agency

FDA Food and drug administration

HA Hyaluronic acid

HCC Hepatocellular carcinoma

LM Laparoscopic myomectomy

PAE Prostatic artery embolization

PVA Polyvinyl alcohol

SCH-58261 4-(Furan-2-yl)-10-(2-phenylethyl)-3,5,6,8,10,11-hexazatricyclo[7.3.0.0]dodeca-1(9),2,4,7,11-pentaen-7-amine

TACE Transarterial chemoembolization

TARE Transarterial radioembolization

TP53 Tumor suppressor gene

UAE Uterine artery embolization

CRediT authorship contribution statement

Andrey N. Kuskov: Writing – review & editing, Validation, Supervision, Project administration, Methodology, Conceptualization. **Pavel P. Kulikov:** Writing – review & editing, Supervision, Resources, Project administration. **Pavel A. Yudaev:** Writing – original draft, Visualization, Validation, Software, Methodology. **Anton S. Tupikov:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software. **Ekaterina A. Lesovaya:** Writing – review & editing, Supervision, Project administration. **Anton A. Keskinov:** Supervision, Resources, Project administration. **Alexander A. Artyukhov:** Methodology, Conceptualization.

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Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

No data was used for the research described in the article.

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